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Yanai et al.

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(54) **MEDIUM PROCESSING APPARATUS,
IMAGE FORMING SYSTEM, MEDIUM
PROCESSING METHOD, AND
NON-TRANSITORY RECORDING MEDIUM**

(71) Applicants: **Yuta Yanai**, Kanagawa (JP); **Yuji
Susaki**, Kanagawa (JP)

(72) Inventors: **Yuta Yanai**, Kanagawa (JP); **Yuji
Susaki**, Kanagawa (JP)

(73) Assignee: **Ricoh Company, Ltd.**, Tokyo (JP)

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May 25, 2023 (JP) 2023-086434

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B65H 37/04 (2006.01)

(52) **U.S. Cl.**
CPC ... **B65H 37/04** (2013.01); **B65H 2301/43828**
(2013.01)

(58) **Field of Classification Search**
CPC B65H 2301/43828; B65H 2301/51616;
B65H 2405/1134; B65H 37/04
See application file for complete search history.

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Primary Examiner — Jennifer Bahls

(74) *Attorney, Agent, or Firm* — IPUSA, PLLC

(57) **ABSTRACT**

A medium processing apparatus includes a conveyor, a receptacle, an aligner, a crimp binder, and circuitry. The conveyor conveys a medium. The receptacle holds the medium. The aligner contacts a downstream end of the medium and aligns a position of the medium in a conveyance direction of the medium. The crimp binder faces the downstream end of the medium to press and deform a binding position on media of which the position is aligned, to bind the media. The aligner and the crimp binder are movable in a main scanning direction orthogonal to each of the conveyance direction and a thickness direction of the medium. When determining that the aligner overlaps the binding position when the receptacle is viewed in the thickness direction, the circuitry moves the aligner in the main scanning direction away from the binding position and causes the crimp binder to crimp and bind the media.

13 Claims, 16 Drawing Sheets

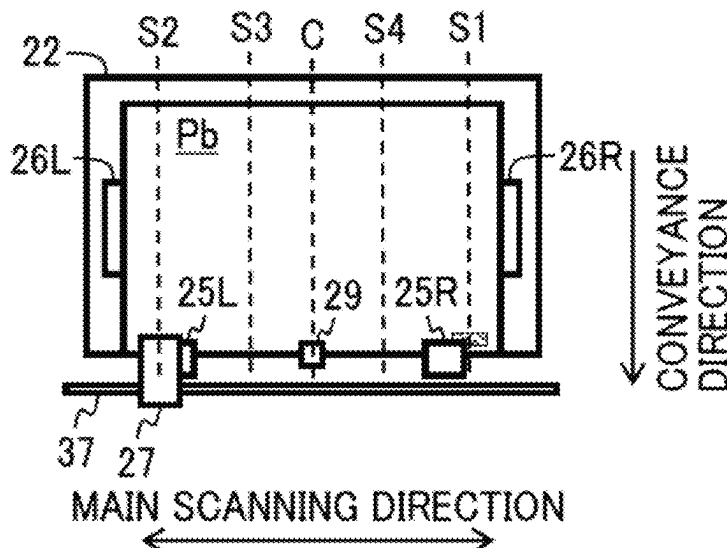


FIG. 1

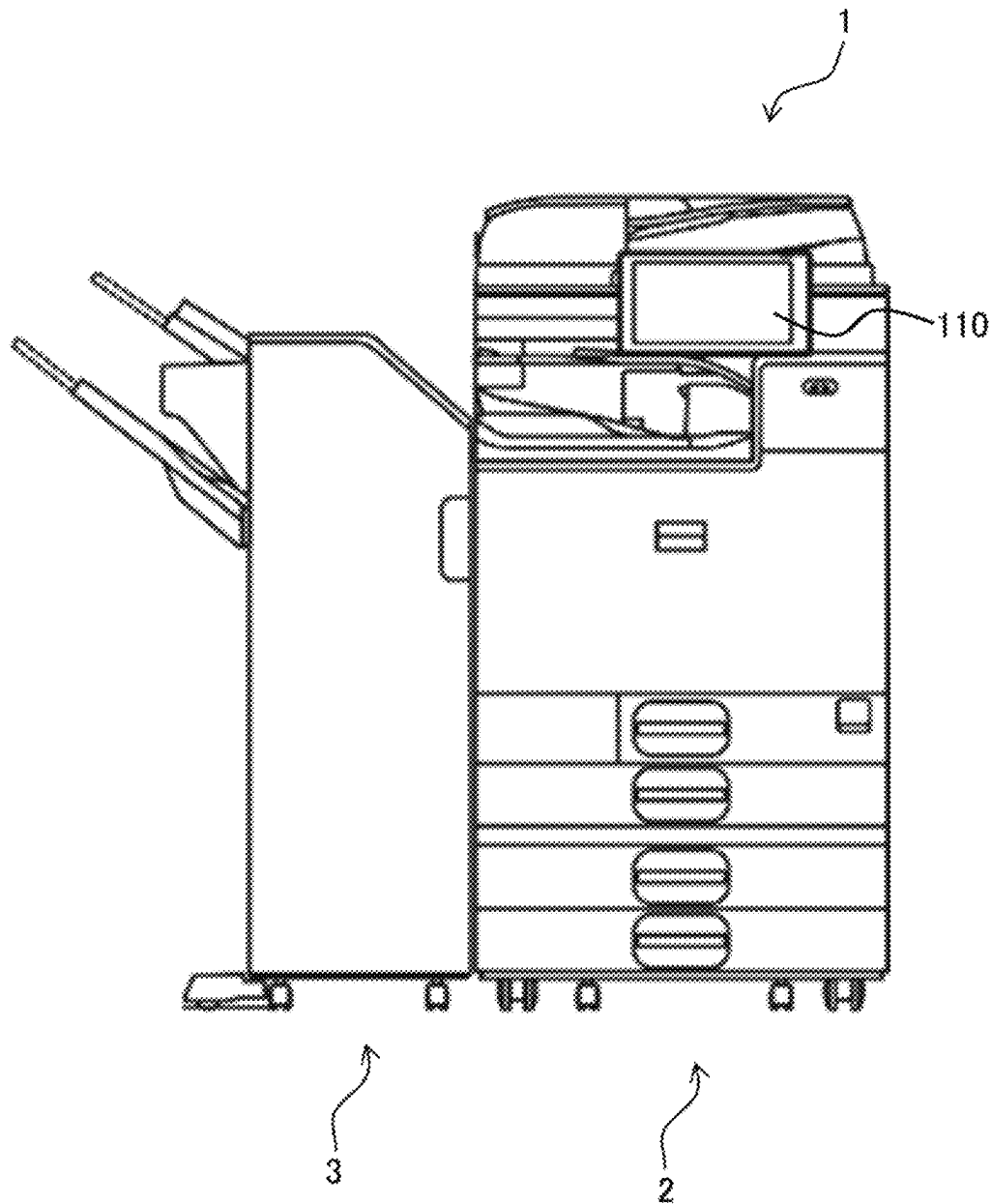


FIG. 2

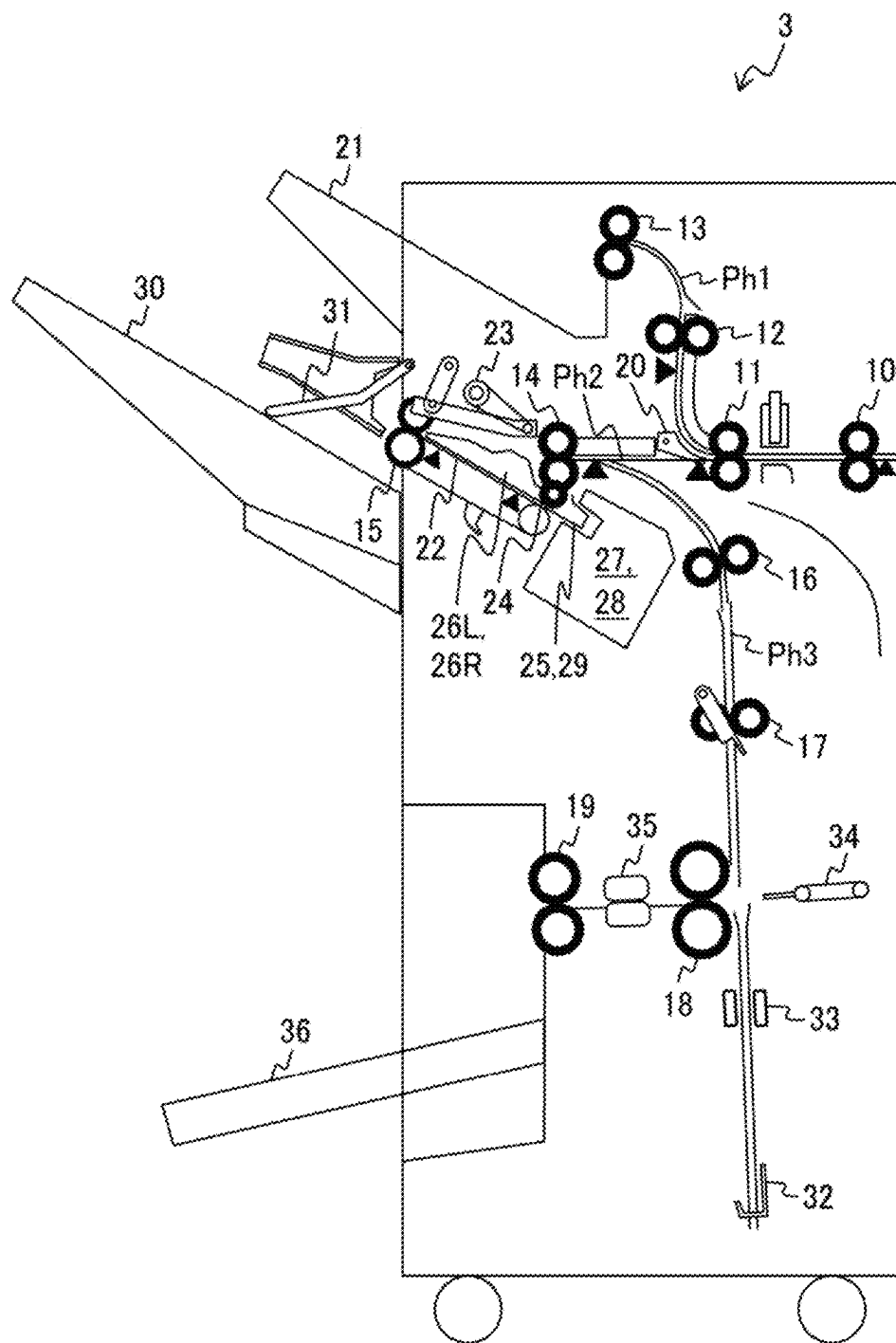


FIG. 4A

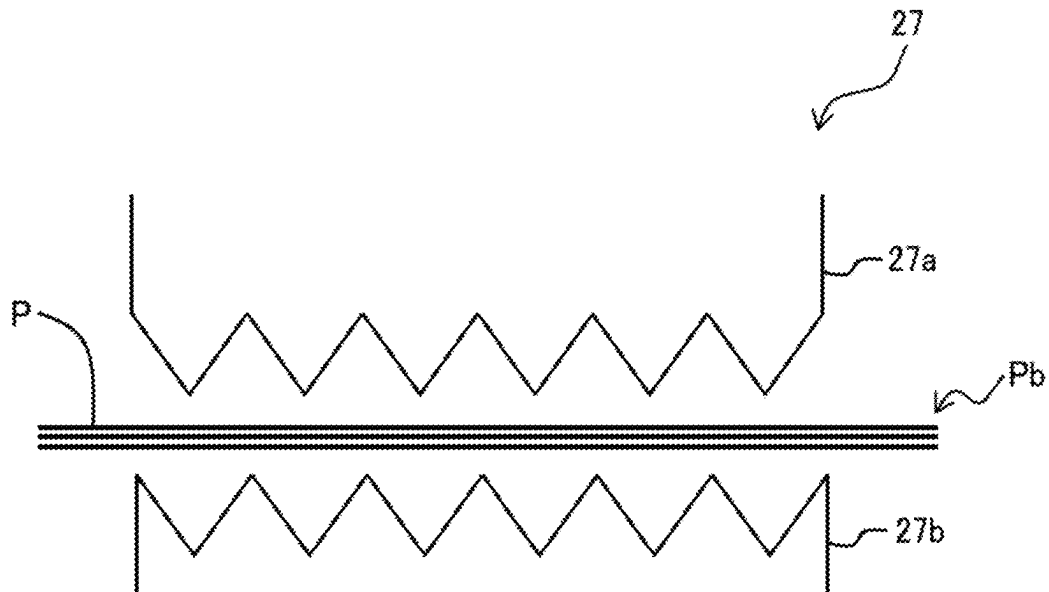


FIG. 4B



FIG. 5

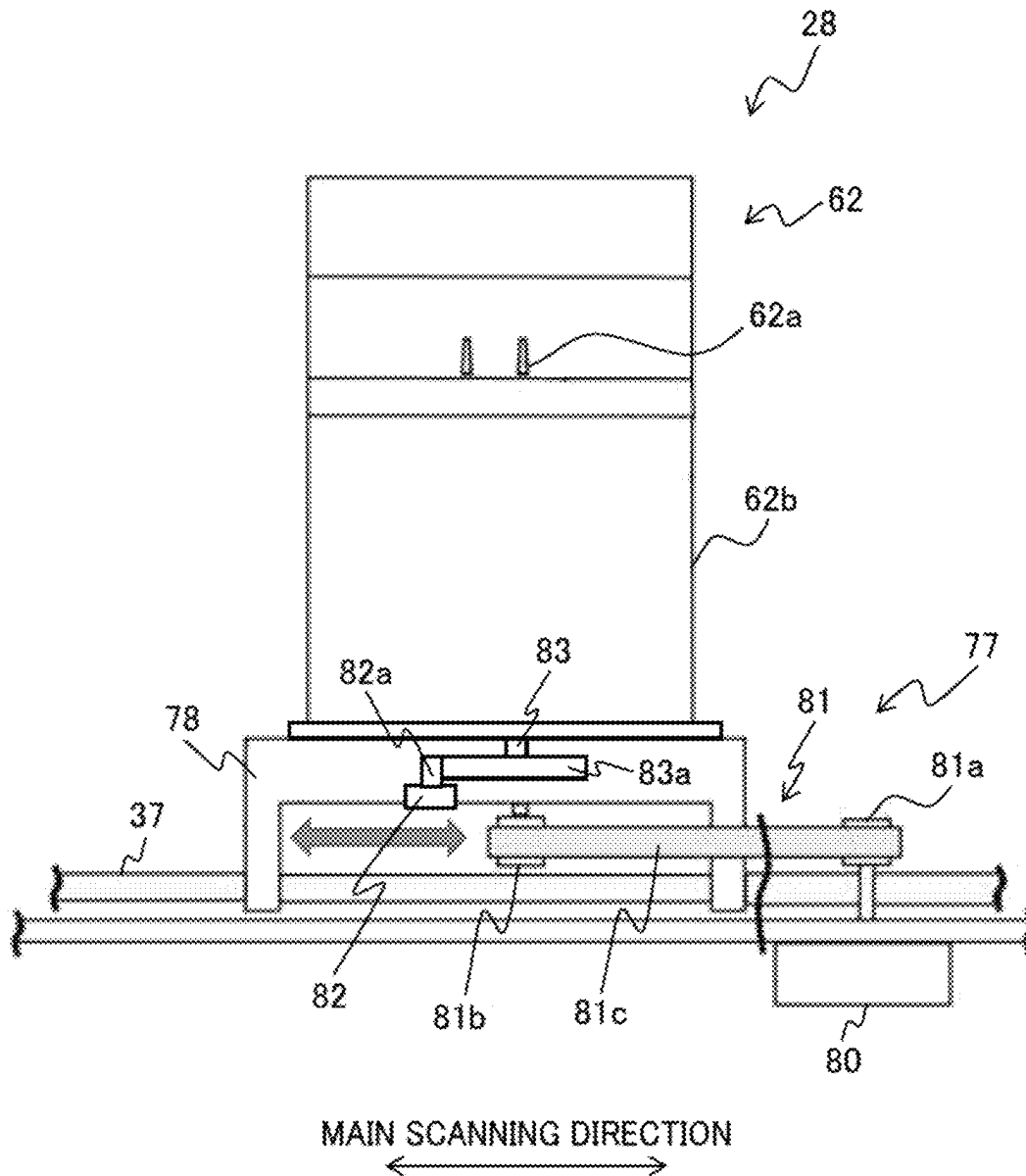


FIG. 6A

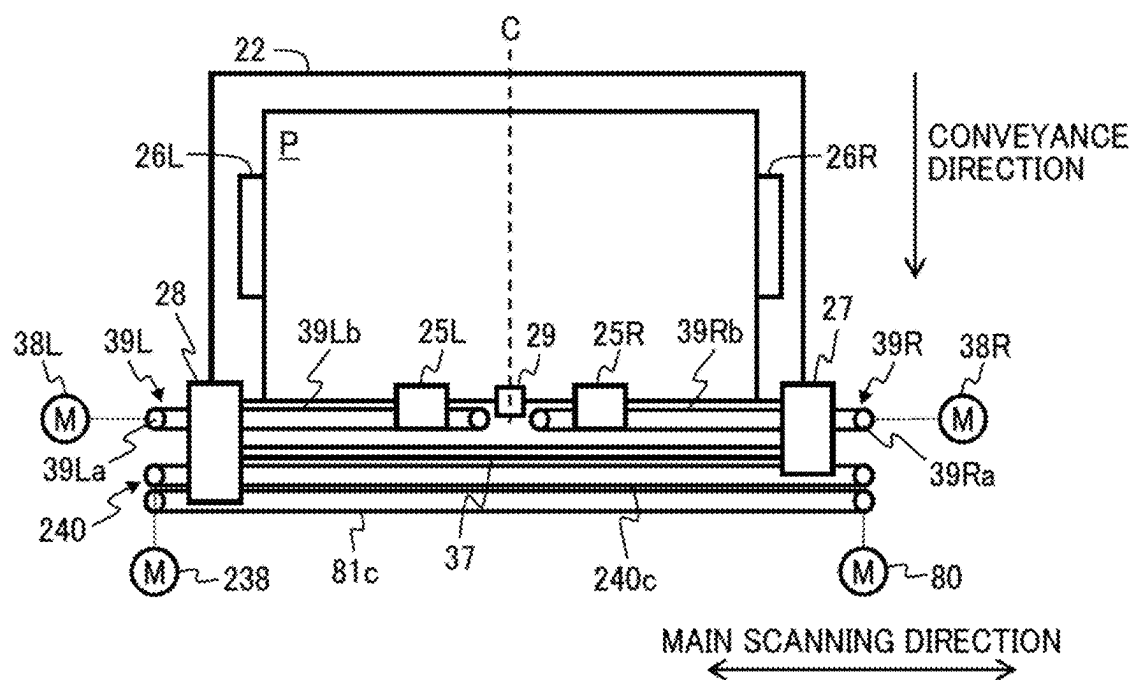


FIG. 6B

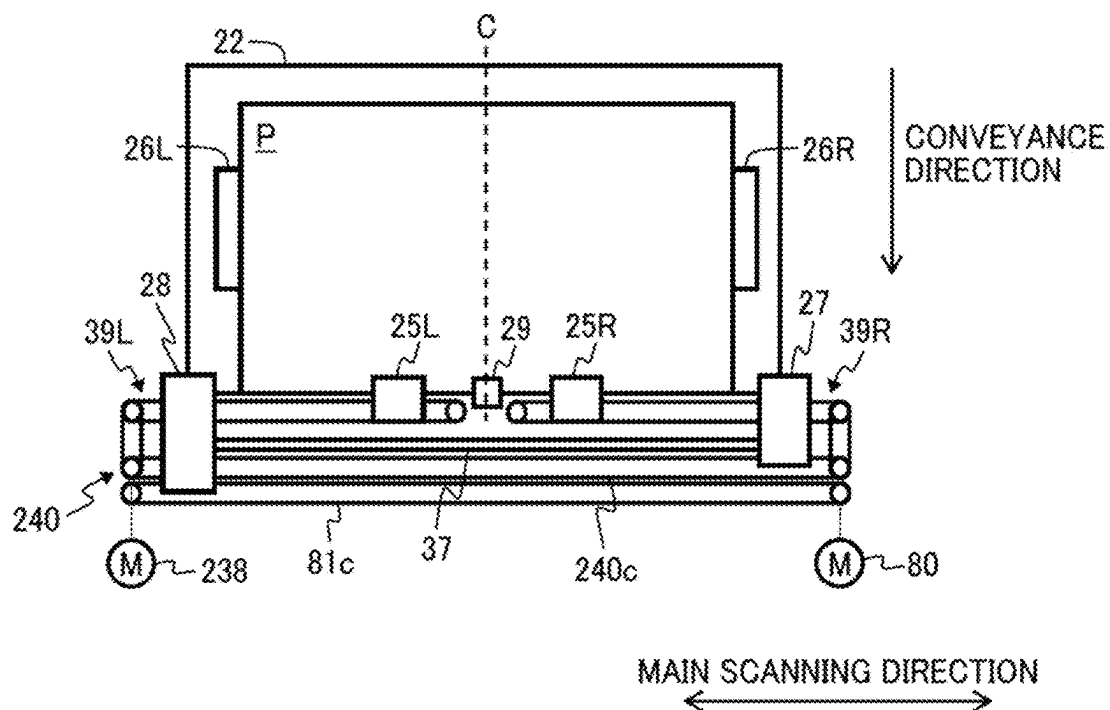


FIG. 7

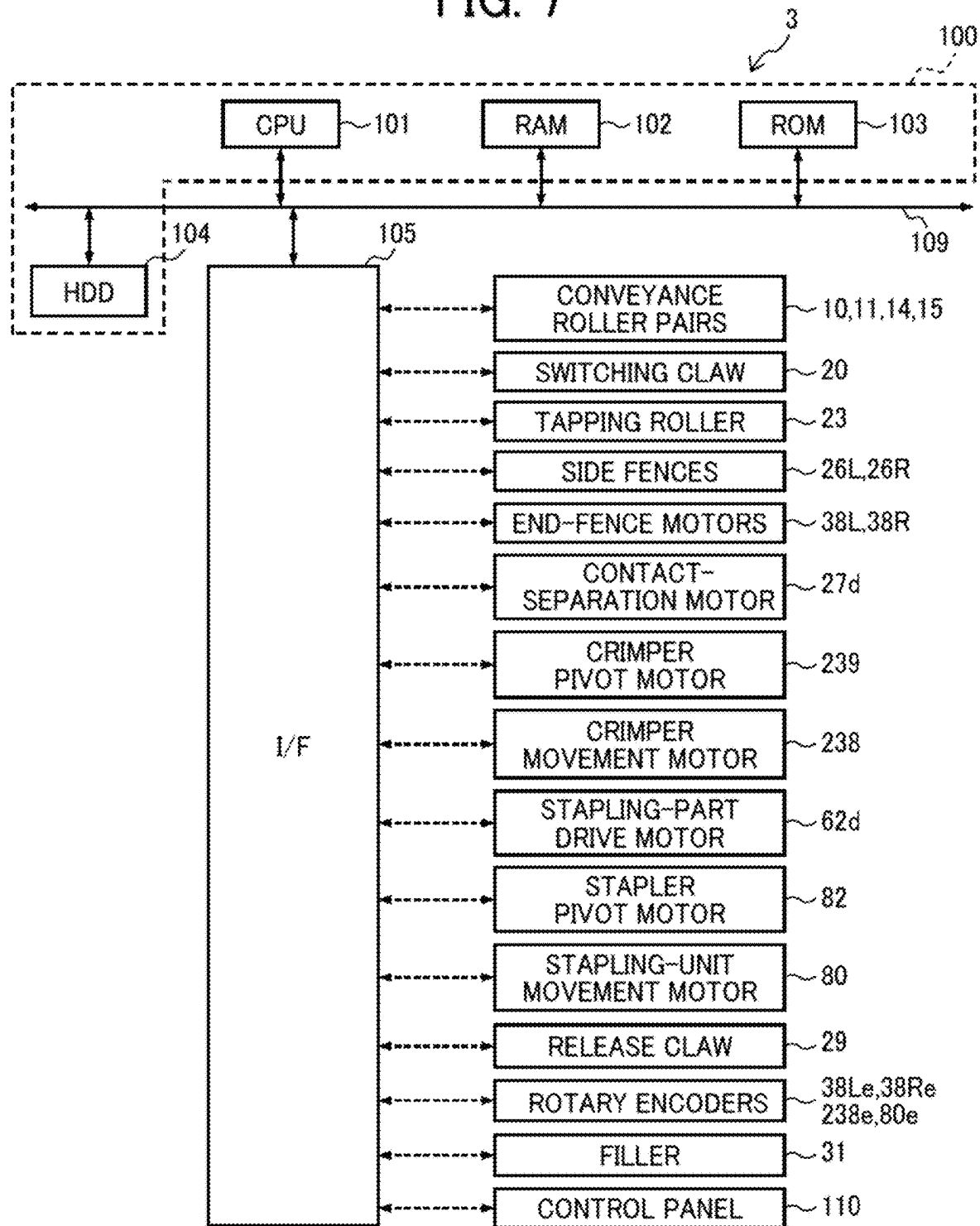


FIG. 8

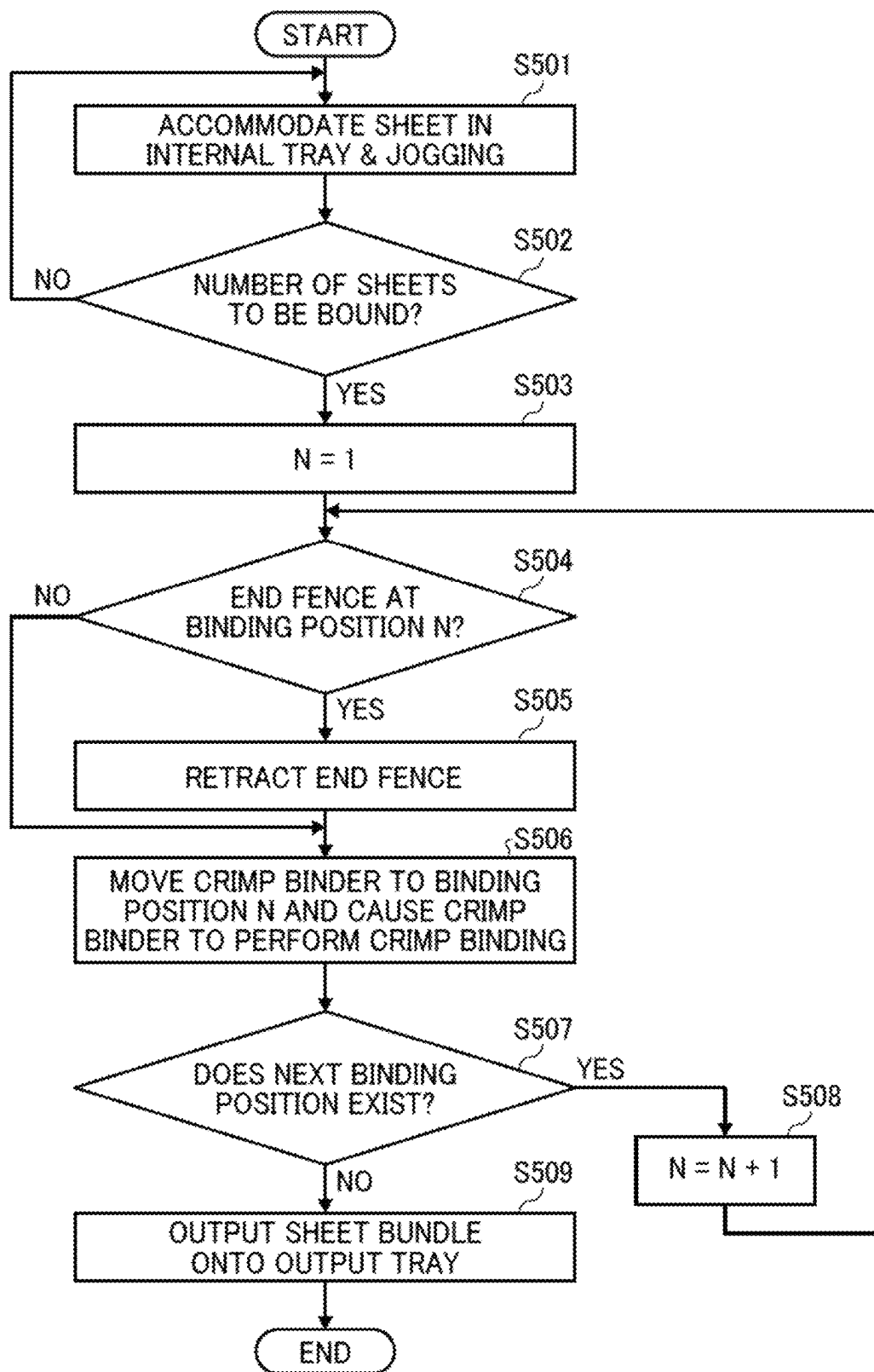


FIG. 9A

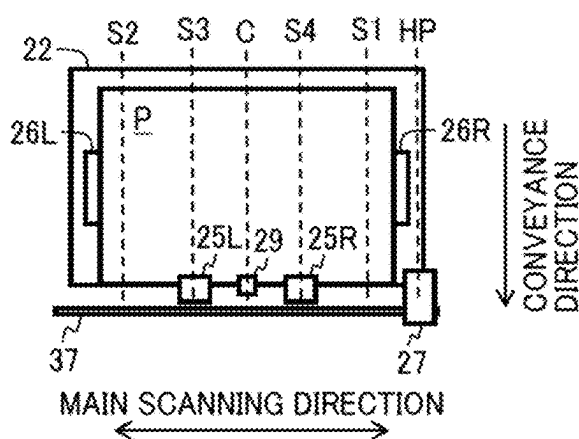


FIG. 9B

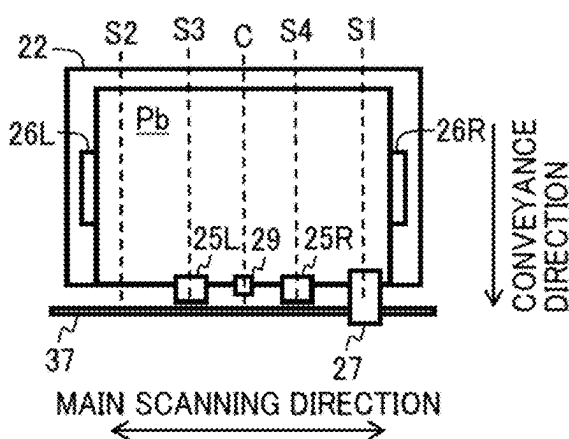


FIG. 9C

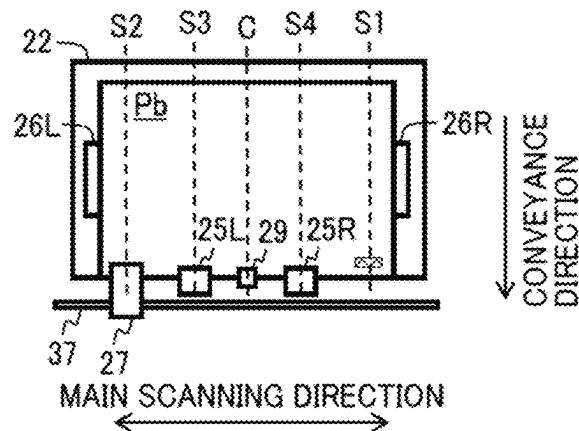


FIG. 9D

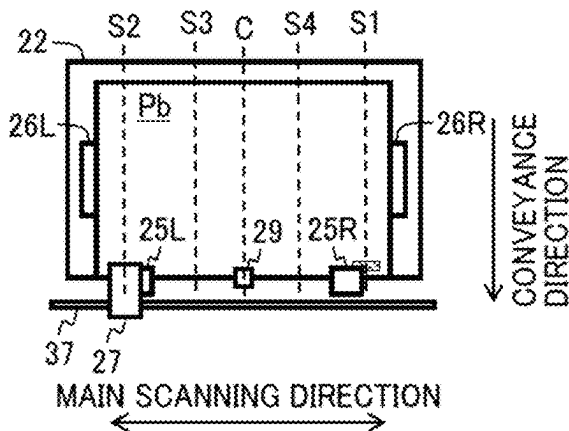


FIG. 9E

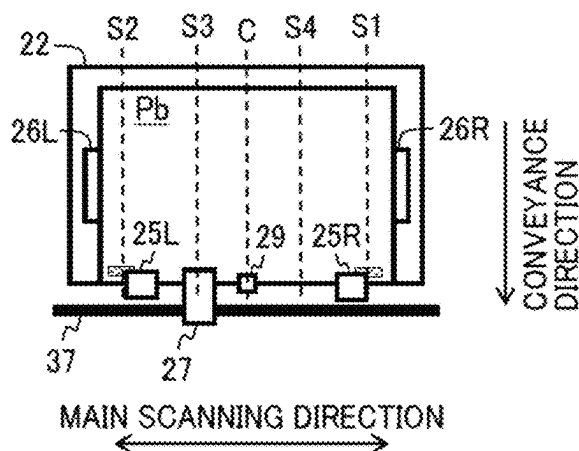


FIG. 9F

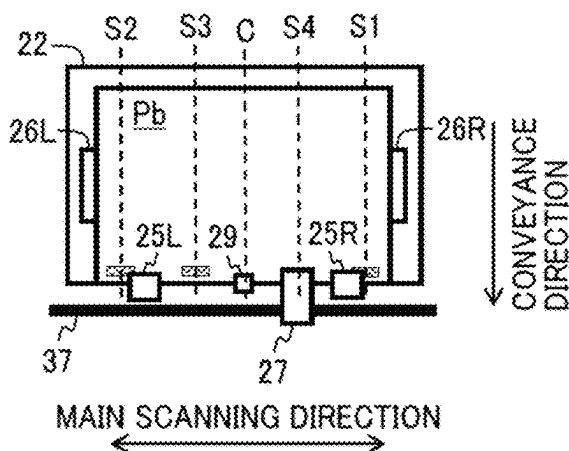


FIG. 10A

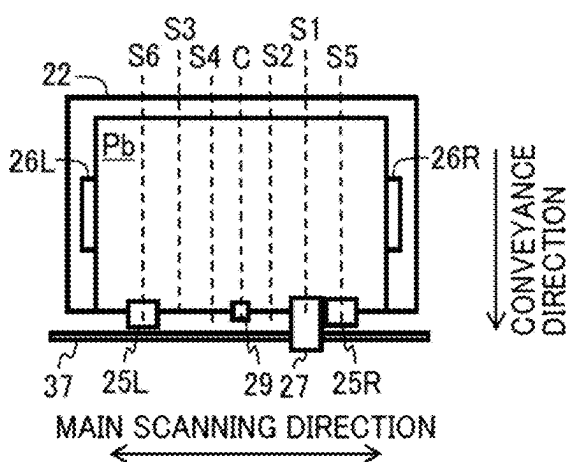


FIG. 10B

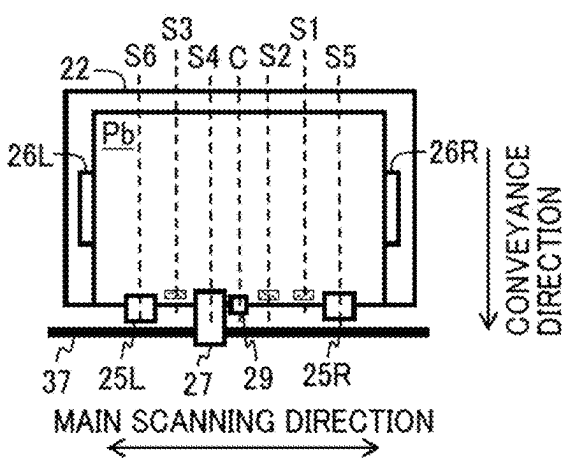


FIG. 10C

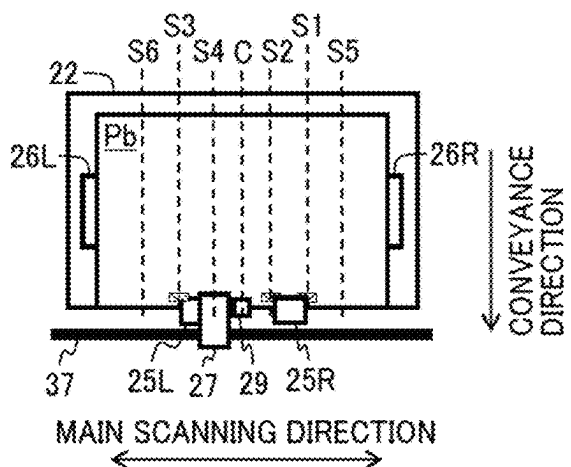


FIG. 10D

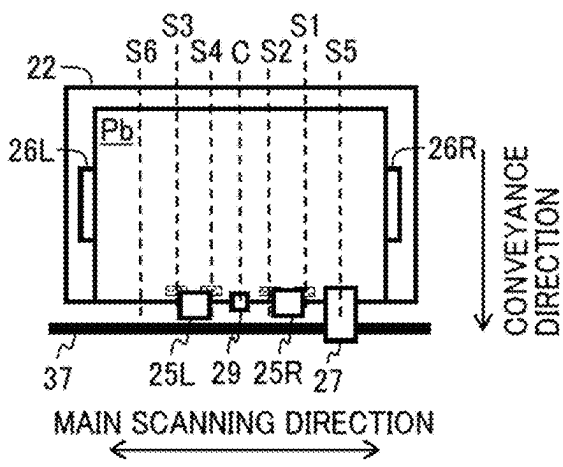


FIG. 10E

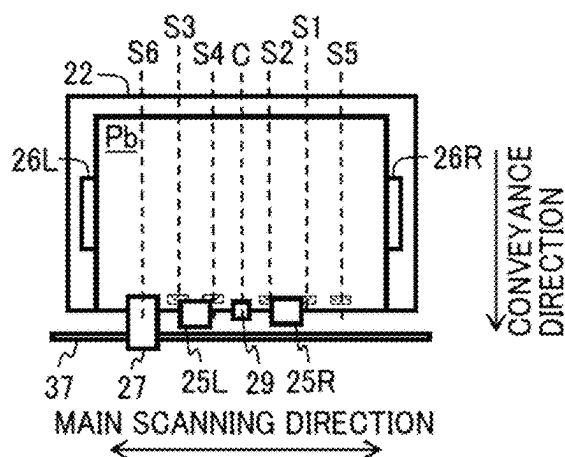


FIG. 11A

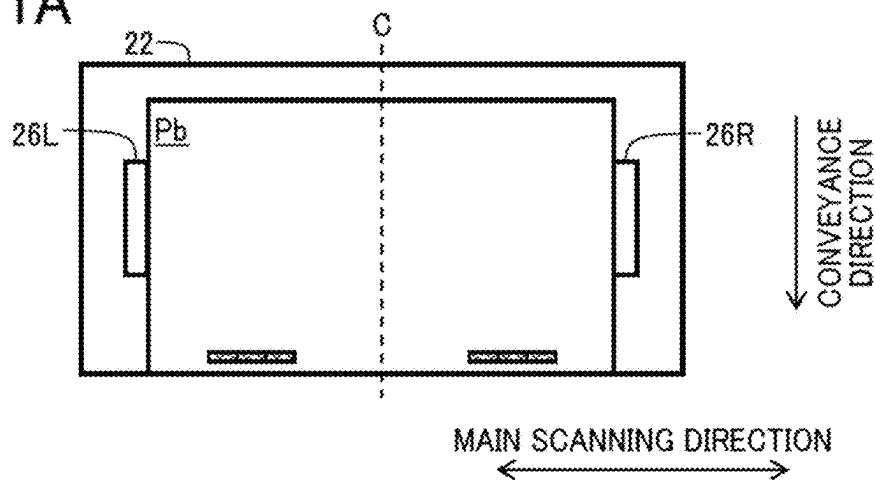


FIG. 11B

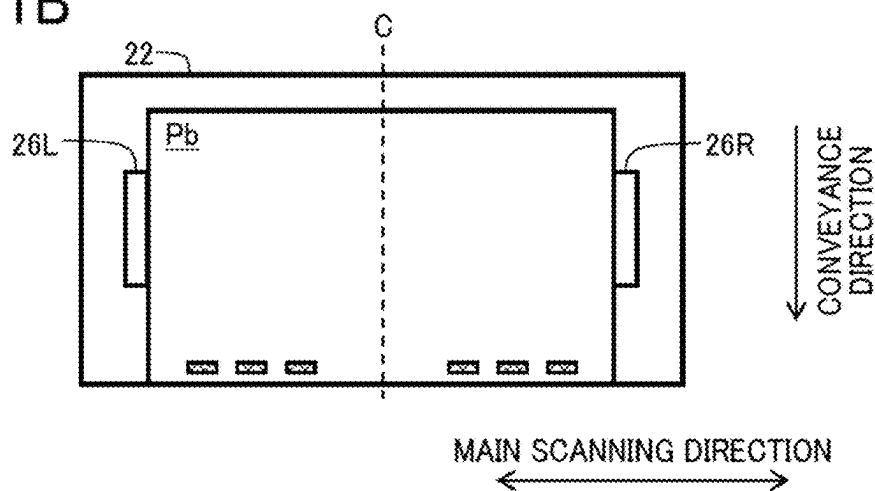


FIG. 11C

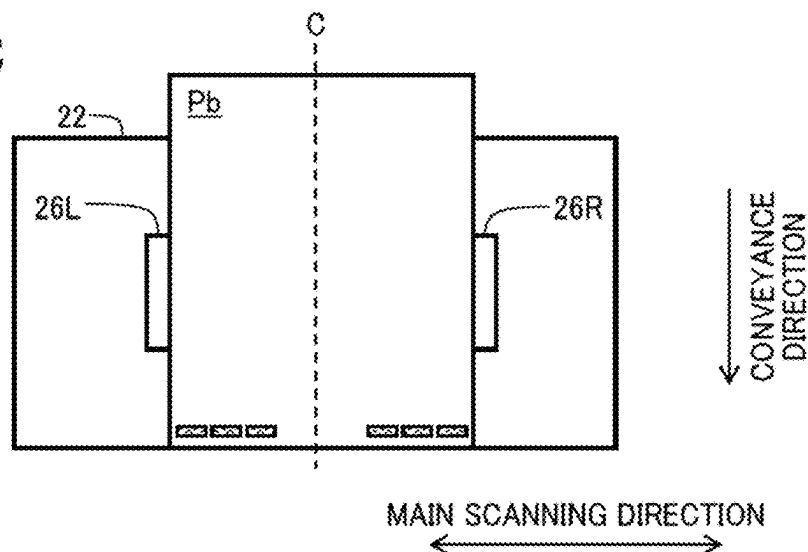


FIG. 12

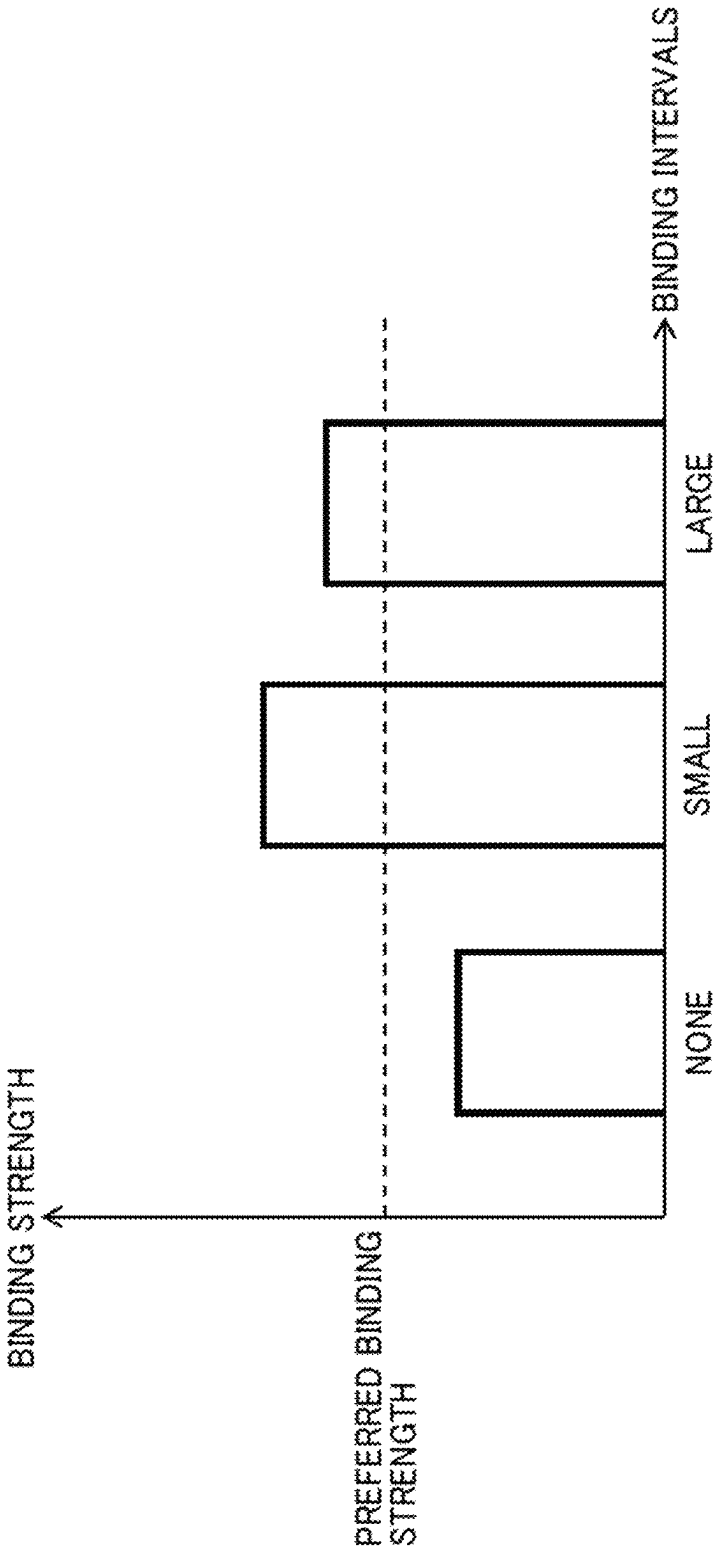


FIG. 13A

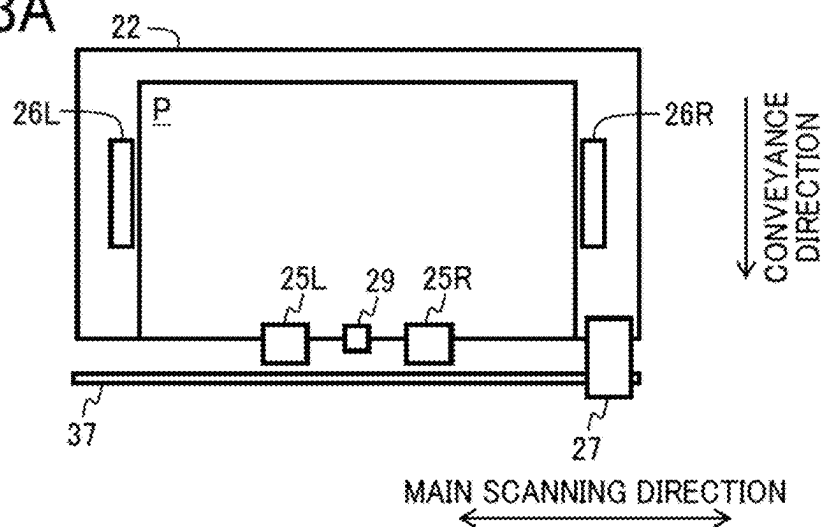


FIG. 13B

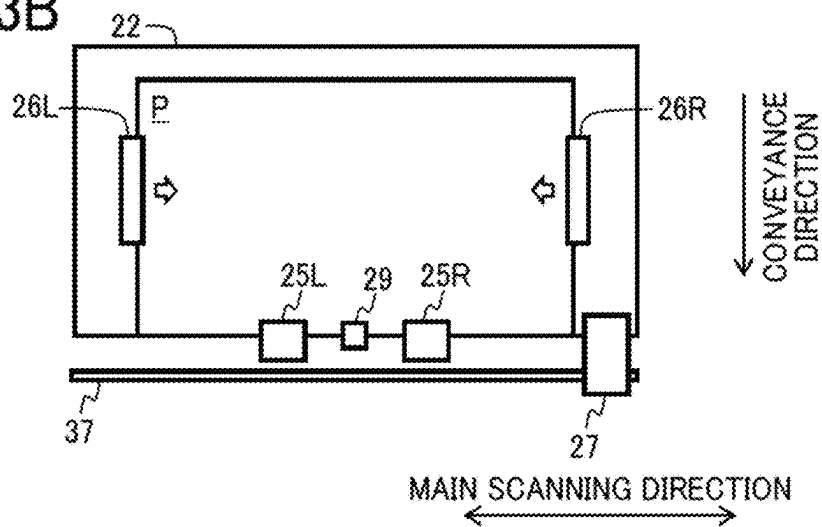


FIG. 13C

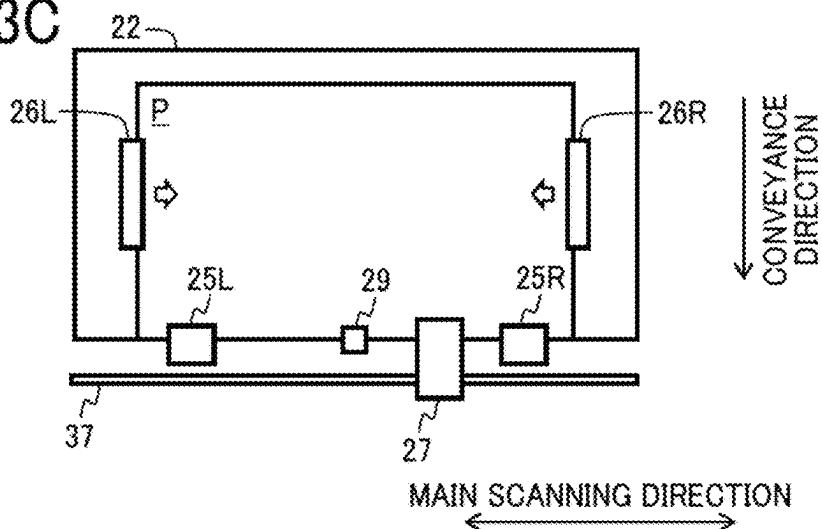


FIG. 14A

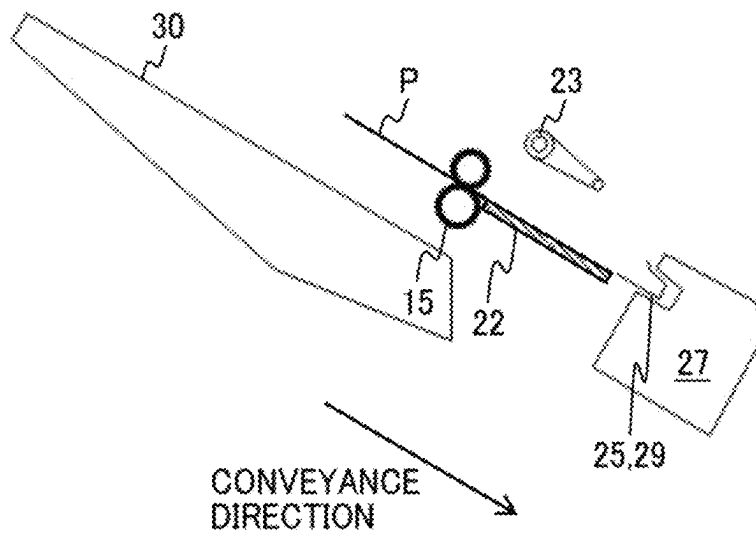


FIG. 14B

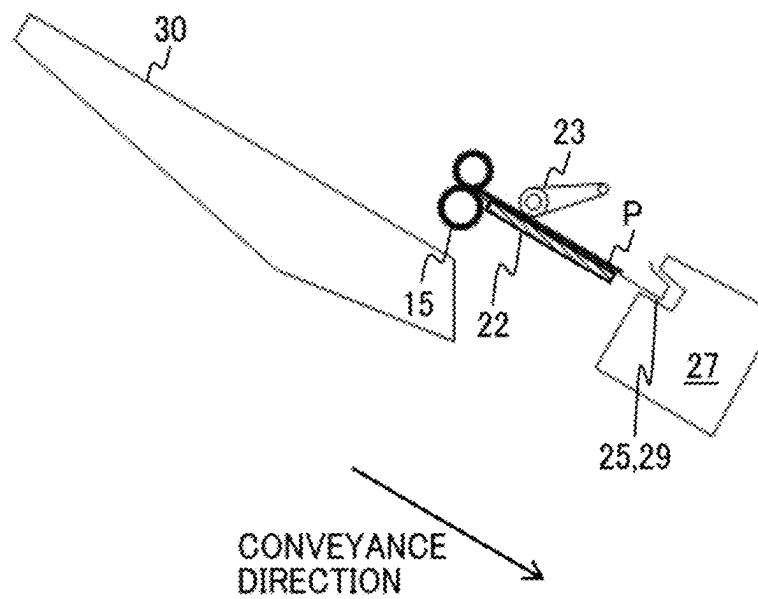


FIG. 15A

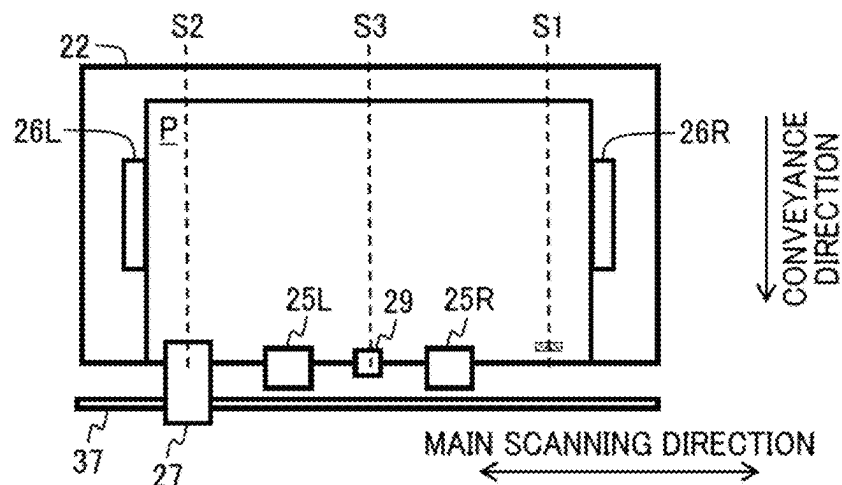


FIG. 15B

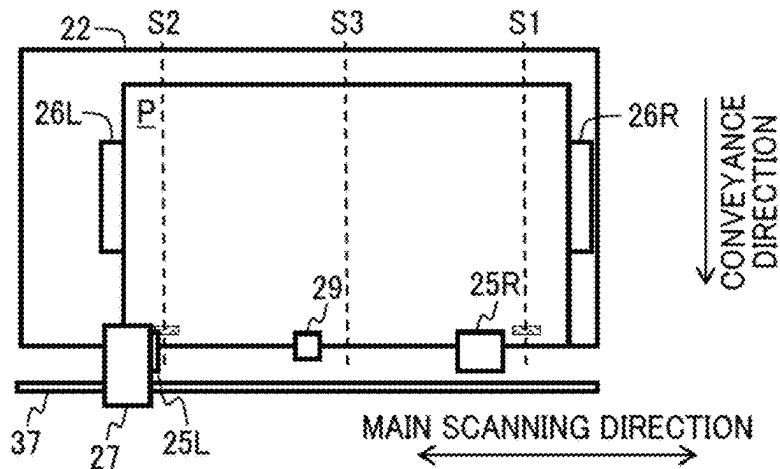


FIG. 15C

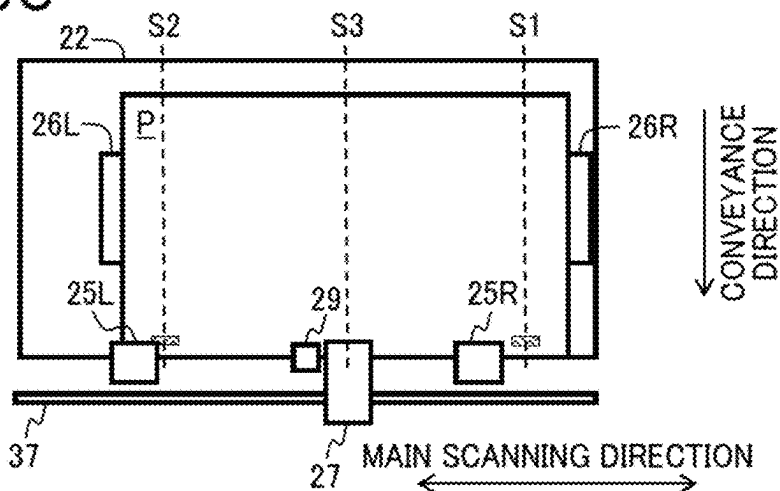


FIG. 16A

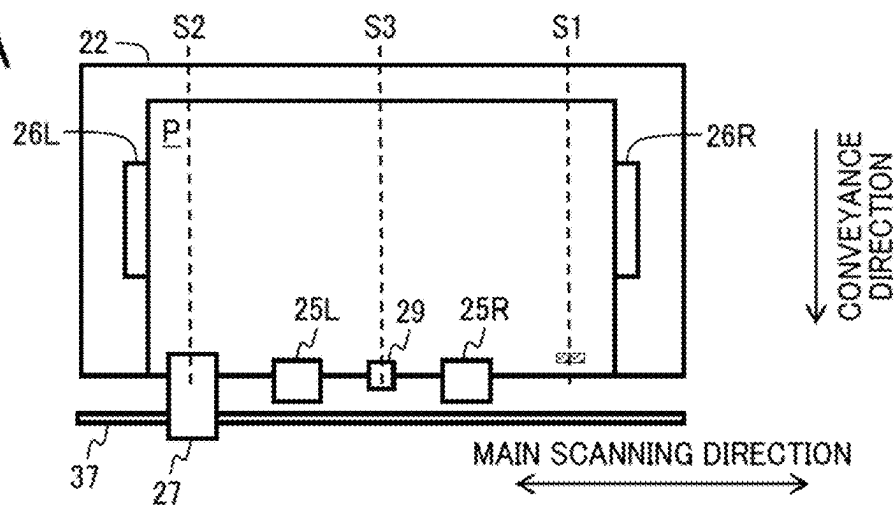


FIG. 16B

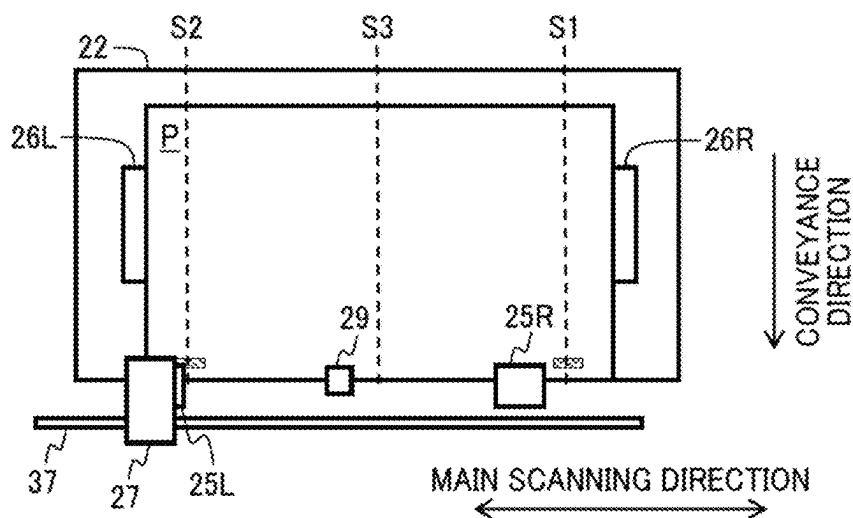
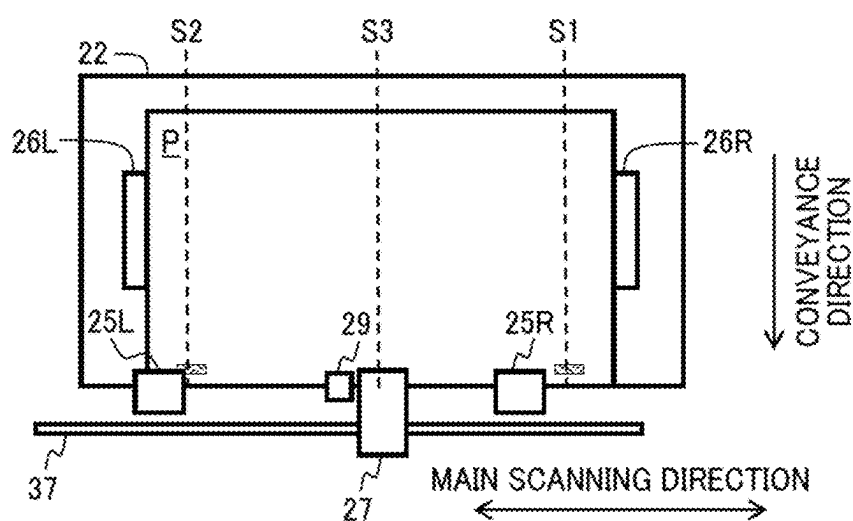


FIG. 16C



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**MEDIUM PROCESSING APPARATUS,
IMAGE FORMING SYSTEM, MEDIUM
PROCESSING METHOD, AND
NON-TRANSITORY RECORDING MEDIUM**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application Nos. 2022-121881, filed on Jul. 29, 2022, and 2023-086434, filed on May 25, 2023, in the Japan Patent Office, the entire disclosure of each of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

Embodiments of the present disclosure relate to a medium processing apparatus, an image forming system, a medium processing method, and a non-transitory recording medium.

Related Art

Medium processing apparatuses are known in the related art that bind, into a bundle, sheet-shaped media on which images are formed by image forming apparatuses. Since sheets of paper are widely known as an example of sheet-shaped media, a “sheet bundle” that is a stack of sheets of paper is used as an example of a bundle of sheet-shaped media in the following description. Some medium processing apparatuses include a crimper that can perform so-called “crimp binding” without metal binding needles from a viewpoint of resource saving and reduction in environmental load. Specifically, the crimper sandwiches a sheet bundle with serrate binding teeth to press and deform the sheet bundle.

SUMMARY

According to an embodiment of the present disclosure, a medium processing apparatus includes a conveyor, a receptacle, a conveyance-direction aligner, a crimp binder, and circuitry. The conveyor conveys a sheet-shaped medium in a conveyance direction. The receptacle holds the medium conveyed by the conveyor. The conveyance-direction aligner contacts a downstream end, in the conveyance direction, of the medium placed on the receptacle and aligns a position, in the conveyance direction, of the medium. The crimp binder is disposed to face the downstream end, in the conveyance direction, of the medium placed on the receptacle to press and deform a binding position on a plurality of media, including the medium, of which the position in the conveyance direction is aligned by the conveyance-direction aligner, to bind the plurality of media. The circuitry controls operations of the conveyor, the conveyance-direction aligner, and the crimp binder. The conveyance-direction aligner and the crimp binder are movable in a main scanning direction orthogonal to each of the conveyance direction and a thickness direction of the medium placed on the receptacle. The circuitry determines whether the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the circuitry determines that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction, the circuitry moves the convey-

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ance-direction aligner in the main scanning direction to a position away from the binding position. The circuitry then causes the crimp binder to crimp and bind the plurality of media.

According to an embodiment of the present disclosure, a novel image forming system includes an image forming apparatus and the medium processing apparatus. The image forming apparatus forms an image on a medium. The medium processing apparatus crimps and binds a plurality of media, including the medium, on each of which the image is formed by the image forming apparatus.

According to an embodiment of the present disclosure, a novel medium processing method includes determining whether a conveyance-direction aligner overlaps a binding position when a receptacle is viewed in a thickness direction of a medium, moving the conveyance-direction aligner in a main scanning direction to a position away from the binding position based on a determination that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction of the medium, the main scanning direction being a direction orthogonal to each of a conveyance direction of the medium and the thickness direction of the medium placed on the receptacle, and moving a crimp binder in the main scanning direction and causing the crimp binder to crimp and bind a plurality of media including the medium.

According to an embodiment of the present disclosure, a novel non-transitory recording medium stores a plurality of instructions which, when executed by one or more processors, causes the processors to perform the medium processing method.

BRIEF DESCRIPTION OF THE DRAWINGS

A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

FIG. 1 is a diagram illustrating the overall configuration of an image forming system according to an embodiment of the present disclosure;

FIG. 2 is a diagram illustrating an internal configuration of a post-processing apparatus in the image forming system of FIG. 1;

FIG. 3 is a schematic view of an upstream side of a crimp binder of the post-processing apparatus of FIG. 2 in a conveyance direction;

FIGS. 4A and 4B are schematic diagrams illustrating a configuration of the crimp binder of FIG. 3;

FIG. 5 is a schematic view of a downstream upstream side of a stapling unit of the post-processing apparatus of FIG. 2 in a conveyance direction;

FIGS. 6A and 6B are views of an internal tray of the post-processing apparatus of FIG. 2 in a thickness direction of a sheet;

FIG. 7 is a block diagram illustrating a hardware configuration of the post-processing apparatus of FIG. 2 to control the post-processing apparatus;

FIG. 8 is a flowchart of a binding process performed by the crimp binder of FIG. 3;

FIGS. 9A to 9F are diagrams illustrating an example of the positions of end fences and the crimp binder of FIG. 3 during the binding process of FIG. 8;

FIGS. 10A to 10E are diagrams illustrating another example of the positions of end fences and the crimp binder of FIG. 3 during the binding process of FIG. 8;

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FIGS. 11A to 11C are diagrams illustrating variations in intervals between binding positions in a main scanning direction;

FIG. 12 is a graph illustrating a relation between a binding strength and intervals between binding positions;

FIGS. 13A to 13C are diagrams illustrating the positions of side fences during a binding process according to a first modification;

FIGS. 14A and 14B are diagrams illustrating the positions of a tapping roller during a binding process according to a second modification;

FIGS. 15A to 15C are diagrams illustrating the positions of side fences during a binding process according to a third modification; and

FIGS. 16A to 16C are diagrams illustrating the positions of an internal tray and side fences during a binding process according to a fourth modification.

The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

Referring now to the drawings, embodiments of the present disclosure are described below.

As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

For the sake of simplicity, like reference numerals are given to identical or corresponding constituent elements such as parts and materials having the same functions, and redundant descriptions thereof are omitted unless otherwise required.

As used herein, the term “connected-coupled” includes both direct connections and connections in which there are one or more intermediate connecting elements.

With reference to the drawings, a description is now given of an image forming system 1 according to an embodiment of the present disclosure.

FIG. 1 is a diagram illustrating the overall configuration of the image forming system 1.

The image forming system 1 has a function of forming an image on a sheet P as a sheet-shaped medium and performing post-processing on the sheet P on which the image is formed. As illustrated in FIG. 1, the image forming system 1 includes an image forming apparatus 2 and a post-processing apparatus 3 serving as a medium processing apparatus.

The image forming apparatus 2 forms an image on the sheet P and outputs the sheet P bearing the image to the post-processing apparatus 3. The image forming apparatus 2 includes a tray that accommodates the sheet P, a conveyor that conveys the sheet P accommodated in the tray, and an image forming device that forms an image on the sheet P conveyed by the conveyor. The image forming device may be an inkjet image forming device that forms an image with

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ink or an electrophotographic image forming device that forms an image with toner. Since the image forming apparatus 2 has a typical configuration, a detailed description of the configuration and functions of the image forming apparatus 2 will be omitted unless otherwise required.

FIG. 2 is a diagram illustrating an internal configuration of the post-processing apparatus 3.

The post-processing apparatus 3 performs post-processing on the sheet P on which an image is formed by the image forming apparatus 2. The post-processing according to the present embodiment is binding as a process to bind the sheets P on each of which an image is formed as a bundle of sheets P. In the following description, the bundle of sheets P may be referred to as a “sheet bundle Pb” serving as a bundle of media. More specifically, the binding according to the present embodiment includes so-called “crimp binding” and “stapling.” The crimp binding is a process to press and deform the sheet bundle Pb to bind the sheet bundle Pb. The stapling is a process to pass a staple through the sheet bundle Pb to bind the sheet bundle Pb. The binding includes edge binding and saddle binding. The edge binding is a process to bind an end (including an edge) of the sheet bundle Pb. The saddle binding is a process to bind the center of the sheet bundle Pb.

The post-processing apparatus 3 includes conveyance roller pairs 10 to 19 serving as conveyors and a switching claw 20. The conveyance roller pairs 10 to 19 convey, inside the post-processing apparatus 3, the sheet P supplied from the image forming apparatus 2. More specifically, the conveyance roller pairs 10 to 13 convey the sheet P along a first conveyance passage Ph1. The conveyance roller pairs 14 and 15 convey the sheet P along a second conveyance passage Ph2. The conveyance roller pairs 16 to 19 convey the sheet P along a third conveyance passage Ph3.

The first conveyance passage Ph1 is a passage extending to a first output tray 21 from a supply port through which the sheet P is supplied from the image forming apparatus 2. The second conveyance passage Ph2 is a passage branching from the first conveyance passage Ph1 between the conveyance roller pairs 11 and 14 in a conveyance direction and extending to a second output tray 30 via an internal tray 22. The third conveyance passage Ph3 is a passage branching from the first conveyance passage Ph1 between the conveyance roller pairs 11 and 14 in the conveyance direction and extending to a third output tray 36.

The switching claw 20 is disposed at a branching position of the first conveyance passage Ph1 and the second conveyance passage Ph2.

The switching claw 20 can be switched between a first position and a second position. The switching claw 20 in the first position guides the sheet P to be output to the first output tray 21 through the first conveyance passage Ph1. The switching claw 20 in the second position guides the sheet P conveyed through the first conveyance passage Ph1 to the second conveyance passage Ph2. When a trailing end of the sheet P entering the second conveyance passage Ph2 passes through the conveyance roller pair 11, the conveyance roller pair 14 is rotated in the reverse direction to guide the sheet P to the third conveyance passage Ph3. The post-processing apparatus 3 further includes a plurality of sensors that detects the positions of the sheet P in the first conveyance passage Ph1, the second conveyance passage Ph2, and the third conveyance passage Ph3. Each of the plurality of sensors is indicated by a black triangle mark in FIG. 2.

The post-processing apparatus 3 includes the first output tray 21. The sheet P output through the first conveyance passage Ph1 is placed on the first output tray 21. Among the

sheets P supplied from the image forming apparatus 2, the sheets P that are not bound are output to the first output tray 21.

The post-processing apparatus 3 further includes the internal tray 22 serving as a receptacle, a tapping roller 23 serving as a contact and separation device, a return roller 24, an end fence 25 serving as a conveyance-direction aligner, side fences 26L and 26R serving as width-direction aligners, a crimp binder 27, a stapling unit 28, a release claw 29 serving as a releaser, the second output tray 30, and a filler 31. The internal tray 22, the tapping roller 23, the return roller 24, the end fence 25, the side fences 26L and 26R, the crimp binder 27, the stapling unit 28, and the release claw 29 perform the edge binding on the sheet bundle Pb constructed of the sheets P conveyed through the second conveyance passage Ph2. Among the sheets P supplied from the image forming apparatus 2, the sheet bundle Pb subjected to the edge binding is output to the second output tray 30.

The “edge binding” includes “parallel binding,” “oblique binding,” and “vertical binding.” The “parallel binding” is a process to bind the sheet bundle Pb along one side of the sheet bundle Pb parallel to a main scanning direction. The “oblique binding” is a process to bind a corner of the sheet bundle Pb. The “vertical binding” is a process to bind the sheet bundle Pb along one side of the sheet bundle Pb parallel to the conveyance direction.

In the following description, a direction in which the sheet P is conveyed from the conveyance roller pair 15 toward the end fence 25 is defined as a “conveyance direction” of the sheet P. In other words, the “conveyance direction” corresponds to a direction in which the sheet P that has been output from the image forming apparatus 2 is moved toward the end fence 25 by the conveyance roller pair 15 after being moved toward the second output tray 30 by, for example, the conveyance roller pair 10. A direction that is orthogonal to each of the conveyance direction and a thickness direction of the sheet P is defined as a “main scanning direction” or a “width direction of the sheet P.”

The internal tray 22 is disposed downstream from the conveyance roller pair 15 in the conveyance direction. The sheets P that are sequentially conveyed through the second conveyance passage Ph2 are temporarily placed on the internal tray 22 serving as a receptacle. The internal tray 22 according to the present embodiment is inclined downward in the conveyance direction. The tapping roller 23 is rotatably held at an end of a rotary arm above the internal tray 22. The tapping roller 23 comes into contact with and separates from the uppermost sheet P placed on the internal tray 22 by the rotation of the rotary arm. The return roller 24 is disposed above the internal tray 22, downstream from the tapping roller 23 in the conveyance direction.

When the rotary arm is pivoted in a direction in which the tapping roller 23 is separated from the internal tray 22, the sheet P conveyed in the conveyance direction by the conveyance roller pair 15 enters the internal tray 22. In this state, when the rotary arm is pivoted in a direction in which the tapping roller 23 approaches the internal tray 22, the tapping roller 23 abuts, from above, against the sheet P conveyed onto the internal tray 22 by the conveyance roller pair 15. The sheet P in contact with the tapping roller 23 is separated from the conveyance roller pair 15 and placed on the internal tray 22. The return roller 24 contacts the upper face of the sheet P placed on the internal tray 22 and rotates to guide the sheet P toward the end fence 25.

The end fence 25 is disposed downstream from the internal tray 22 in the conveyance direction. The end fence 25 is movable in the main scanning direction along the

surface of the sheet P or the sheet bundle Pb placed on the internal tray 22. The end fence 25 contacts a downstream end, in the conveyance direction, of the sheet P or the sheet bundle Pb placed on the internal tray 22 to align the downstream end, in the conveyance direction, of the plurality of sheets P of the sheet bundle Pb. The side fences 26L and 26R are disposed on opposed sides on the internal tray 22 in the main scanning direction. The side fences 26L and 26R are movable in the main scanning direction. The side fences 26L and 26R contact opposed ends, in the main scanning direction, of the sheet P or the sheet bundle Pb placed on the internal tray 22 to align the ends, in the main scanning direction, of the plurality of sheets P of the sheet bundle Pb.

The crimp binder 27 and the stapling unit 28 are disposed downstream from the internal tray 22 in the conveyance direction. The crimp binder 27 and the stapling unit 28 are independently movable in the main scanning direction along the surface of the sheet P the sheet bundle Pb placed on the internal tray 22. The crimp binder 27 and the stapling unit 28 perform the edge binding on an end of the sheet bundle Pb aligned by the end fence 25 and the side fences 26L and 26R.

More specifically, the crimp binder 27 sandwiches the binding position on the sheet bundle Pb placed on the internal tray 22 with senate binding teeth from both sides in the thickness direction. As a result, the binding position on the sheet bundle Pb is pressed and deformed. Thus, the sheet bundle Pb is crimped and bound.

FIG. 3 is a schematic view of an upstream side of the crimp binder 27 in the conveyance direction.

As illustrated in FIG. 3, a guide shaft 37 extends in the main scanning direction at a position downstream from the internal tray 22 in the conveyance direction. The crimp binder 27 is moved in the main scanning direction along the surface of the sheet bundle Pb placed on the internal tray 22, in other words, along the guide shaft 37, by a driving force that is transmitted from a crimper movement motor 238 by a driving force transmission assembly 240 including pulleys 240a and 240b and a timing belt 240c. A crimper shaft 340 provided with a drive transmission gear 340a is fixed to a bottom face of a crimping frame 27c that holds the components of the crimp binder 27 such as upper crimping teeth 27a and lower crimping teeth 27b. The crimper shaft 340 and the drive transmission gear 340a are held by a base 48 on which the crimping frame 27c is disposed, so as to be rotatable in the forward and reverse directions. The drive transmission gear 340a meshes with an output gear 239a of a crimper pivot motor 239. The crimp binder 27 is rotated in the forward and reverse directions on the base 48 about the crimper shaft 340 extending in the thickness direction of the sheet P placed on the internal tray 22, by a driving force transmitted from the crimper pivot motor 239 to the crimper shaft 340 via the output gear 239a and the drive transmission gear 340a. The guide shaft 37, the crimper movement motor 238, the crimper pivot motor 239, the crimper shaft 340, and the driving force transmission assembly 240 construct a driving assembly of the crimp binder 27.

As illustrated in FIGS. 4A and 4B, the crimp binder 27 includes a pair of binding teeth (i.e., the upper crimping teeth 27a and the lower crimping teeth 27b). The upper crimping teeth 27a and the lower crimping teeth 27b are disposed to face each other in the thickness direction of the sheet bundle Pb so as to sandwich the sheet bundle Pb placed on the internal tray 22. The upper crimping teeth 27a and the lower crimping teeth 27b have respective serrate faces facing each other. The senate face of each of the upper crimping teeth 27a and the lower crimping teeth 27b

includes concave portions and convex portions alternately formed. The concave portions and the convex portions of the upper crimping teeth **27a** are shifted from those of the lower crimping teeth **27b** such that the upper crimping teeth **27a** are engaged with the lower crimping teeth **27b**. The upper crimping teeth **27a** and the lower crimping teeth **27b** are brought into contact with and separated from each other by a driving force of a contact-separation motor **27d** illustrated in FIG. 7.

In the process of supplying the sheets P of the sheet bundle Pb to the internal tray **22**, the upper crimping teeth **27a** and the lower crimping teeth **27b** are apart from each other as illustrated in FIG. 4A. When all the sheets P of the sheet bundle Pb are placed on the internal tray **22**, the upper crimping teeth **27a** and the lower crimping teeth **27b** are engaged with each other to press and deform the sheet bundle Pb in the thickness direction as illustrated in FIG. 4B. As a result, the sheet bundle Pb that has been placed on the internal tray **22** is crimped and bound. The sheet bundle Pb thus crimped and bound is output to the second output tray by the conveyance roller pair **15**.

The configuration of the crimp binder **27** as a crimping assembly is not limited to the configuration of a moving assembly exemplified in the present embodiment, provided that the upper crimping teeth **27a** and the lower crimping teeth **27b** of the crimping assembly are engaged with each other. For example, the crimping assembly may be a crimping assembly disclosed in Japanese Patent No. 6057167 or its corresponding U.S. Patent Application Publication No. 2014-0219747, which is hereby incorporated by reference as though disclosed herein in its entirety. In this case, the crimping assembly brings the upper crimping teeth **27a** and the lower crimping teeth **27b** into contact with each other and separates the upper crimping teeth **27a** and the lower crimping teeth **27b** from each other with a link assembly and a driving source that simply rotates forward or that rotates forward and backward. Alternatively, the crimping assembly may employ a linear motion system to linearly bring the upper crimping teeth **27a** and the lower crimping teeth **27b** into contact with each other and separate the upper crimping teeth **27a** and the lower crimping teeth **27b** from each other with a screw assembly that converts the forward and backward rotational motions of a driving source into linear reciprocating motion.

Now, a detailed description is given of the stapling unit **28**.

The stapling unit **28** executes stapling or a stapling process. Specifically, the stapling unit **28** passes a staple through a binding position on the sheet bundle Pb placed on the internal tray **22** to bind the sheet bundle Pb.

FIG. 5 is a schematic view of a downstream side of the stapling unit **28** in the conveyance direction.

The stapling unit **28** includes a stapler **62** that binds the sheet bundle Pb with a staple or staples. The stapler **62** is disposed downstream from the internal tray **22** in the conveyance direction and apart from the crimp binder **27** in the main scanning direction.

The stapler **62** has a configuration for performing so-called “stapling” (i.e., stapling process) to bind the sheet bundle Pb with a staple or staples. More specifically, the stapler **62** includes a stapling-part drive motor **62d** illustrated in FIG. 7. The stapling-part drive motor **62d** drives a stapling part **62a**. A driving force of the stapling-part drive motor **62d** causes a staple loaded in the stapling part **62a** to pass through the sheet bundle Pb. Thus, the stapling part **62a**

binds the sheet bundle Pb. Since the stapler **62** has a typical configuration, a detailed description thereof will be omitted unless otherwise required.

As illustrated in FIG. 5, the stapling unit **28** includes a stapling-unit movement assembly **77**.

The stapling-unit movement assembly **77** moves the stapling unit **28** in the main scanning direction along the downstream end, in the conveyance direction, of the sheet P or the sheet bundle Pb placed on the internal tray **22**.

The stapling-unit movement assembly **77** includes, for example, a base **78**, the guide shaft **37**, a stapling-unit movement motor **80**, and a driving force transmission assembly **81**. The driving force transmission assembly **81** transmits a driving force of the stapling-unit movement motor **80** to the base **78** via pulleys **81a** and **81b** and a timing belt **81c**. A stapler shaft **83** provided with a drive transmission gear **83a** is fixed to a bottom face of a stapling frame **62b** that holds the components of the stapler **62**. The stapler shaft **83** and the drive transmission gear **83a** are held by the base **78** on which the stapling frame **62b** is disposed, so as to be rotatable in the forward and reverse directions. The drive transmission gear **83a** meshes with an output gear **82a** of a stapler pivot motor **82**. The stapler **62** can be rotated in the forward and reverse directions about the stapler shaft **83** on the base **78** by a driving force transmitted from the stapler pivot motor **82** to the stapler shaft **83** via the output gear **82a** and the drive transmission gear **83a**.

The crimp binder **27** and the stapling unit **28** are supported by the common guide shaft **37**. The driving force transmission assembly **240** and the stapling-unit movement assembly **77** move the crimp binder **27** and the stapling unit **28** in the main scanning direction along the common guide shaft **37**. The driving force transmission assembly **240** and the stapling-unit movement assembly **77** can independently move the crimp binder **27** and the stapling unit **28**.

The release claw **29** is disposed downstream from the internal tray **22** in the conveyance direction. The position, in the main scanning direction, of the release claw **29** is fixed. The release claw **29** is movable upstream in the conveyance direction (i.e., toward the conveyance roller pair **15**) in contact with the downstream end, in the conveyance direction, of the sheet bundle Pb placed on the internal tray **22**. The release claw **29** moving upstream in the conveyance direction moves the sheet bundle Pb subjected to the edge binding upstream in the conveyance direction along the internal tray **22**. As a result, the sheet bundle Pb subjected to the edge binding is released from the internal tray **22** and enters between rollers of the conveyance roller pair **15**. Then, the conveyance roller pair **15** outputs the sheet bundle Pb subjected to the edge binding to the second output tray **30**.

The second output tray **30** is disposed on an outer side face of a housing of the post-processing apparatus **3** such that the second output tray **30** can vertically move.

The filler **31** is rotatable above the second output tray **30**. An end of the filler **31** contacts the sheet bundle Pb placed on the second output tray **30**. The filler **31** detects that the height (thickness) of the sheet bundle Pb stacked on the second output tray **30** reaches a threshold. The filler **31** then outputs the detection result to a controller **100** illustrated in FIG. 7. A detailed description of the controller **100** is deferred. When the filler **31** detects that the height of the sheet bundle Pb reaches the threshold, the controller **100** lowers the second output tray **30** by a given amount.

The post-processing apparatus **3** further includes an end fence **32**, a saddle binder **33**, a sheet folding blade **34**, a hole punch **35**, and the third output tray **36**. The end fence **32**, the

saddle binder **33**, and the sheet folding blade **34** perform the saddle binding on the sheet bundle Pb constructed of the sheets P that are conveyed through the third conveyance passage Ph3. Among the sheets P supplied from the image forming apparatus **2**, the sheet bundle Pb subjected to the saddle binding is output to the third output tray **36**.

The end fence **32** aligns the positions, in a direction in which the sheets P are conveyed, of the sheets P that are sequentially conveyed through the third conveyance passage Ph3. The end fence **32** is movable between a binding position where the end fence **32** causes the center of the sheet bundle Pb to face the saddle binder **33** and a folding position where the end fence **32** causes the center of the sheet bundle Pb to face the sheet folding blade **34**. The saddle binder **33** binds the center of the sheet bundle Pb aligned by the end fence **32** at the binding position. The sheet folding blade **34** folds, in half, the sheet bundle Pb placed on the end fence **32** at the folding position and causes the conveyance roller pair **18** to sandwich the sheet bundle Pb. The conveyance roller pairs **18** and **19** output the sheet bundle Pb subjected to the saddle binding to the third output tray **36**. The hole punch **35** punches a through hole in the sheet bundle Pb that is conveyed by the conveyance roller pairs **15** and **19**.

Each of FIGS. **6A** and **6B** is a view of the internal tray **22** in the thickness direction of the sheet P.

As illustrated in FIGS. **6A** and **6B**, the post-processing apparatus **3** according to the present embodiment includes a symmetrical pair of end fences **25L** and **25R**. The end fences **25L** and **25R** are arranged opposite each other across the release claw **29** in the main scanning direction. In other words, the end fences **25L** and **25R** are arranged opposite each other across a center position C, in the main scanning direction, of the sheet bundle Pb placed on the internal tray **22**.

The end fences **25L** and **25R** are disposed at equal distances from the release claw **29**. As described above, the crimp binder **27** and the stapling unit **28** are supported by the guide shaft **37** extending in the main scanning direction at a position downstream from the internal tray **22** in the conveyance direction, such that the crimp binder **27** and the stapling unit **28** can independently move in the main scanning direction. In FIGS. **6A** and **6B** and subsequent figures, illustration of the stapling unit **28** may be omitted.

The end fences **25L** and **25R** according to the present embodiment move between respective proximate positions illustrated in FIGS. **9A** to **9C** and respective distant positions illustrated in FIGS. **9D** to **9F**. The interval between the end fences **25L** and **25R** in the main scanning direction is greater at the respective distant positions than at the respective proximate positions. In other words, the distant position is farther from the center position C than the proximate position. Note that the distance from the center position C to the end fence **25L** at the proximate position is equal to the distance from the center position C to the end fence **25R** at the proximate position. Similarly, the distance from the center position C to the end fence **25L** at the distant position is equal to the distance from the center position C to the end fence **25R** at the distant position. Thus, the end fences **25L** and **25R** move in the main scanning direction while maintaining the same distance from the center position C. In other words, the interval between the end fences **25L** and **25R** in the main scanning direction increases or decreases. The positions to which the end fences **25L** and **25R** can move are not limited to the positions illustrated in FIGS. **9A** to **9F**.

On the other hand, the crimp binder **27** according to the present embodiment moves between a standby position HP

illustrated in FIG. **9A** and positions where the crimp binder **27** faces binding positions **S1** to **S4** as illustrated in FIGS. **9B** to **9F**. The standby position HP is away in one of the main scanning directions from the sheet P placed on the internal tray **22**. For example, in FIG. **9A**, the standby position HP is distanced to the right of the sheet P in the main scanning direction. The binding positions **S1** to **S4** are positions on the sheet bundle Pb placed on the internal tray **22**. However, the specific positions of the binding positions **S1** to **S4** are not limited to the positions illustrated in FIGS. **9A** and **9F**. The binding positions **S1** to **S4** may be positions in the main scanning direction at the downstream end, in the conveyance direction, of the sheet bundle Pb.

As an example, as illustrated in FIG. **6A**, the end fences **25L** and **25R** are independently moved in the main scanning direction by a driving force transmitted from end-fence motors **38L** and **38R** serving as first driving sources through a driving force transmission assemblies **39L** and **39R**. The driving force transmission assembly **39L** includes, for example, pulleys **39La** and a timing belt **39Lb**. Similarly, the driving force transmission assembly **39R** includes, for example, pulleys **39Ra** and a timing belt **39Rb**. The position, in the main scanning direction, of the end fence **25L** is detected by a sensor disposed within the moving range of the end fence **25L**, a rotary encoder **38Le** (see FIG. **7**) of the end-fence motor **38L**, or a combination thereof. Similarly, the position of the end fence **25R** in the main scanning direction is detected by a sensor disposed within the moving range of the end fence **25R**, a rotary encoder **38Re** (see FIG. **7**) of the end-fence motor **38R**, or a combination thereof. The readings indicating the detected positions of the end fences **25L** and **25R** are provided to the controller **100**.

As described above, the crimp binder **27** is moved in the main scanning direction by the driving force that is transmitted from the crimper movement motor **238** by the driving force transmission assembly **240** including the pulleys **240a** and **240b** and the timing belt **240c**. On the other hand, the stapling unit **28** is moved in the main scanning direction by the driving force that is transmitted from the stapling-unit movement motor **80** by the driving force transmission assembly **81** including the pulleys **81a** and **81b** and the timing belt **81c**. The positions, in the main scanning direction, of the crimp binder **27** and the stapling unit **28** are detected by a rotary encoder **238e** (see FIG. **7**) of the crimper movement motor **238** and a rotary encoder **80e** (see FIG. **7**) of the stapling-unit movement motor **80**, respectively. Readings indicating the detected positions of the crimp binder **27** and the stapling unit **28** are provided to the controller **100**.

As another example, as illustrated in FIG. **6B**, the driving force transmission assemblies **39L** and **39R** and the driving force transmission assembly **240** may be coupled to each other. The end fences **25L** and **25R** and the crimp binder **27** may be moved in the main scanning direction in conjunction with each other by, for example, the driving force transmitted from the crimper movement motor **238** as a common driving source through the driving force transmission assemblies **39L** and **39R** and the driving force transmission assembly **240**. In this case, the end-fence motors **38L** and **38R** illustrated in FIG. **6A** may be omitted.

FIG. **7** is a block diagram illustrating a hardware configuration of the post-processing apparatus **3**.

As illustrated in FIG. **7**, the post-processing apparatus **3** includes a central processing unit (CPU) **101**, a random access memory (RAM) **102**, a read only memory (ROM) **103**, a hard disk drive (HDD) **104**, and an interface (I/F) **105**. The CPU **101**, the RAM **102**, the ROM **103**, the HDD **104**, and the I/F **105** are connected to each other via a common

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bus 109. The CPU 101 is an arithmetic unit and controls the overall operation of the post-processing apparatus 3. The RAM 102 is a volatile storage medium that allows data to be read and written at high speed. The CPU 101 uses the RAM 102 as a work area for data processing. The ROM 103 is a read-only non-volatile storage medium that stores programs such as firmware. The HDD 104 is a non-volatile storage medium that allows data to be read and written and has a relatively large storage capacity. The HDD 104 stores, for example, an operating system (OS), various control programs, and application programs.

By an arithmetic function of the CPU 101, the post-processing apparatus 3 processes, for example, a control program stored in the ROM 103 and an information processing program (application program) loaded into the RAM 102 from a storage medium such as the HDD 104. Such processing configures a software controller including various functional modules of the post-processing apparatus 3. The software controller that is thus configured cooperates with hardware resources of the post-processing apparatus 3 to construct functional blocks that implement functions of the post-processing apparatus 3. In other words, the CPU 101, the RAM 102, the ROM 103, and the HDD 104 construct the controller 100 that controls the operation of the post-processing apparatus 3.

The I/F 105 is an interface that connects the conveyance roller pairs 10, 11, 14, and the switching claw 20, the tapping roller 23, the side fences 26L and 26R, the end-fence motors 38L and 38R, the contact-separation motor 27d, the crimper pivot motor 239, the crimper movement motor 238, the stapling-part drive motor 62d, the stapler pivot motor 82, the stapling-unit movement motor 80, the release claw 29, the rotary encoders 38Le, 38Re, 238e, and 80e, the filler 31, and a control panel 110 to the common bus 109.

The controller 100 controls, via the I/F 105, the operations of the conveyance roller pairs 10, 11, 14, and 15, the switching claw 20, the tapping roller 23, the side fences 26L and 26R, the end-fence motors 38L and 38R, the contact-separation motor 27d, the crimper pivot motor 239, the crimper movement motor 238, the stapling-part drive motor 62d, the stapler pivot motor 82, the stapling-unit movement motor 80, and the release claw 29. On the other hand, the controller 100 acquires detection results from the rotary encoders 38Le, 38Re, 238e, and 80e and the filler 31. Although FIG. 7 illustrates the components related to the stapling unit 28 and the crimp binder 27 that executes the edge binding, the components related to the saddle binder 33 that executes the saddle binding are controlled by the controller 100 like the components related to the stapling unit 28 and the crimp binder 27 that executes the edge binding.

As illustrated in FIG. 1, the image forming apparatus 2 includes the control panel 110. The control panel 110 includes an operation unit that receives instructions from a user and a display serving as a notifier that notifies the user of information. The operation unit includes, for example, hard keys and a touch panel superimposed on the display. The control panel 110 acquires information from the operator through the operation unit and provides information to the operator through the display. Note that a specific example of the notifier is not limited to the display and may be a light emitting diode (LED) lamp or a speaker. The post-processing apparatus 3 may include the control panel 110 like the control panel 110 described above.

FIG. 8 is a flowchart of a binding process.

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FIGS. 9A to 9F are diagrams illustrating an example of the positions of the end fences 25L and 25R and the crimp binder 27 during the binding process.

For example, the controller 100 starts the binding process illustrated in FIG. 8 when the controller 100 acquires an instruction to execute the binding process from the image forming apparatus 2. In the following description, the instruction to execute the binding process may be referred to as a "binding command." The binding command includes, for example, the binding position on the sheet bundle Pb in the main scanning direction, the number of binding positions on the sheet bundle Pb, and the number of sheets P of the sheet bundle Pb. In the following description, the number of sheets P of the sheet bundle Pb may be referred to as the "number of sheets to be bound." As illustrated in FIG. 9A, at the start of the binding process, the end fences 25L and 25R are at the respective proximate positions while and the crimp binder 27 is at the standby position HP.

In step S501, the controller 100 rotates the conveyance roller pairs 10, 11, 14, and 15 to accommodate, in the internal tray 22, the sheet P on which an image is formed by the image forming apparatus 2. The controller 100 also moves the side fences 26L and 26R to align the position, in the main scanning direction, of the sheet P or the sheet bundle Pb placed on the internal tray 22. In short, the controller 100 performs so-called jogging.

Subsequently, in step S502, the controller 100 determines whether the number of sheets P accommodated in the internal tray 22 has reached the number of sheets to be bound, which is instructed by the binding command.

When the controller 100 determines that the number of sheets P accommodated in the internal tray 22 has not reached the number of sheets to be bound (NO in step S502), the controller 100 executes the operation of step S501 again. On the other hand, when the controller 100 determines that the number of sheets P accommodated in the internal tray 22 has reached the number of sheets to be bound (YES in step S502), in step S503, the controller 100 initializes a variable N stored in the HDD 104. In other words, in step S503, the controller 100 sets the variable N to 1.

Subsequently, in step S504, the controller 100 determines whether the end fence 25L or 25R overlaps a binding position N when the internal tray 22 is viewed in the thickness direction. In the present embodiment, as illustrated in FIGS. 9A to 9F, the binding positions S1, S2, S3, and S4 are crimped and bound. The order in which the binding positions S1 to S4 are crimped and bound is not particularly limited. However, the binding position that overlaps neither the end fence 25L nor 25R is preferably crimped and bound first.

When the controller 100 determines that neither the end fence 25L nor 25R overlaps the binding position S1 (NO in step S504), in step S506, the controller 100 causes the crimp binder 27 to crimp and bind the binding position S1 without executing the operation of step S505 (i.e., without moving the end fence 25L or 25R). In other words, as illustrated in FIG. 9B the controller 100 drives the crimper movement motor 238 to move the crimp binder 27 to the position where the crimp binder 27 faces the binding position S1 and causes the crimp binder 27 to sandwich the binding position S1 on the sheet bundle Pb between the upper crimping teeth 27a and the lower crimping teeth 27b to press and deform the binding position S1.

Subsequently, in step S507 the controller 100 determines whether any binding position that is yet to be crimped and bound exists.

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When the controller 100 determines that a binding position that is yet to be crimped and bound exists (YES in step S507), in step S508, the controller 100 increases the variable N by 1. Then, the controller 100 performs the operation of step S504 again.

Specifically, in step S504, the controller 100 determines whether the end fence 25L or 25R overlaps the binding position S2.

When the controller 100 determines that neither the end fence 25L nor 25R overlaps the binding position S2 (NO in step S504), in step S506, the controller 100 causes the crimp binder 27 to crimp and bind the binding position S2 that overlaps neither the end fence 25L nor 25R as illustrated in FIG. 9C.

Thereafter, in step S507, the controller 100 determines again whether any binding position that is yet to be crimped and bound exists.

When the controller 100 determines that a binding position that is yet to be crimped and bound exists (YES in step S507), in step S508, the controller 100 increases the variable N by 1. Then, the controller 100 performs the operation of step S504 again.

Specifically, in step S504, the controller 100 determines whether the end fence 25L or 25R overlaps the binding position S3.

When the controller 100 determines that the end fence 25L or 25R overlaps the binding position S3 (YES in step S504), in step S505, the controller 100 drives the end-fence motors 38L and 38R to move the end fences 25L and 25R to the respective distant positions as illustrated in FIG. 9D. In other words, in step S505, the controller 100 moves the end fences 25L and 25R in the main scanning direction to respective positions away from the binding position S3.

After moving the end fences 25L and 25R, in step S506, the controller 100 causes the crimp binder 27 to crimp and bind the binding position S3 as illustrated in FIG. 9E.

Thereafter, in step S507, the controller 100 determines again whether any binding position that is yet to be crimped and bound exists.

When the controller 100 determines that a binding position that is yet to be crimped and bound exists (YES in step S507), in step S508, the controller 100 increases the variable N by 1. Then, the controller 100 performs the operation of step S504 again.

Specifically, in step S504, the controller 100 determines whether the end fence 25L or 25R overlaps the binding position S4.

When the controller 100 determines that neither the end fence 25L nor 25R overlaps the binding position S4 (NO in step S504), in step S506, the controller 100 causes the crimp binder 27 to crimp and bind the binding position S4 that overlaps neither the end fence 25L nor 25R as illustrated in FIG. 9F.

When the controller 100 determines that all the binding positions S1 to S4 are crimped and bound (NO in step S507), in step S509, the controller 100 outputs the sheet bundle Pb thus crimped and bound to the second output tray 30. More specifically, the controller 100 moves the release claw 29 upstream in the conveyance direction so that the rollers of the conveyance roller pair 15 sandwich the sheet bundle Pb. The controller 100 then rotates the conveyance roller pair 15 to output the sheet bundle Pb to the second output tray 30. The controller 100 also moves the end fences 25L and 25R and the crimp binder 27 to the respective initial positions illustrated in FIG. 9A.

The number of binding positions is not limited to the number of binding positions illustrated in FIGS. 9A to 9F.

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The respective initial positions of the end fences 25L and 25R are not limited to the respective proximate positions.

FIGS. 10A to 10E are diagrams illustrating another example of the positions of the end fences 25L and 25R and the crimp binder 27 during the binding process.

In FIGS. 10A to 10E, six binding positions S1 to S6 are crimped and bound. The end fences 25L and 25R are at the respective distant positions at the start of the binding process.

First, as illustrated in FIGS. 10A and 10B, the controller 100 causes the crimp binder 27 to sequentially crimp and bind the binding positions S1 to S4 that overlap neither the end fence 25L nor 25R. The controller 100 then moves the end fences 25L and 25R to the respective proximate positions away from the binding positions S5 and S6 as illustrated in FIG. 10C. The controller 100 then causes the crimp binder 27 to sequentially crimp and bind the binding positions S5 and S6 that overlap neither the end fence 25L nor 25R as illustrated in FIGS. 10D and 10E.

Preferably, the binding positions are symmetrical with respect to the center position C as illustrated in FIGS. 9A to 10E. The controller 100 retracts the end fences 25L and 25R preferably after causing the crimp binder 27 to crimp and bind at least one binding position on each side of the center position C in the main scanning direction among the plurality of binding positions. The intervals between the plurality of binding positions in the main scanning direction is preferably changed depending on, for example, the number of binding positions or the size of the sheet P in the main scanning direction.

FIGS. 11A to 11C are diagrams illustrating variations in the intervals between the plurality of binding positions in the main scanning direction.

FIG. 12 is a graph illustrating a relation between the binding strength and the intervals between the binding positions.

As illustrated in FIG. 11A, when the intervals are 0 mm between the adjacent binding positions in the main scanning direction, the binding strength is lower than a preferred binding strength as indicated by “none” in the axis representing the binding intervals in FIG. 12. In other words, the plurality of binding positions are preferably arranged at given intervals in the main scanning direction.

Even when the sheets P in the same size are conveyed, the widths of the sheets P in the main scanning direction vary between a case where the sheet P is conveyed with a longitudinal direction of the sheet P in the main scanning direction (i.e., long edge feed [LEF] illustrated in FIG. 11B) and a case where the sheet P is conveyed with a short direction of the sheet P in the main scanning direction (i.e., short edge feed [SEF] illustrated in FIG. 11C). When the sheets P in different sizes (for example, A4 and B5) are conveyed, the widths of the sheets P in the main scanning direction vary. To handle such a situation, the controller 100 may change the intervals between the plurality of binding positions in the main scanning direction depending on the width of the sheet P in the main scanning direction.

Now, a description is given of two example ways to change the arrangement of the binding positions from the arrangement illustrated in FIG. 11B to the arrangement illustrated in FIG. 11C.

As a first example, the intervals between the binding positions are narrowed with the binding positions closest to the center position C fixed. As a second example, the binding positions are shifted to the center position C while maintaining the intervals between the binding positions.

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As illustrated in FIG. 12, when the intervals between the plurality of binding positions in the main scanning direction are equal to or greater than 5 mm as indicated by “large” in the axis representing the binding intervals in FIG. 12, the binding strength decreases compared to a case where the intervals are less than 5 mm as indicated by “small” in the axis representing the binding intervals in FIG. 12. For this reason, the intervals between the plurality of binding positions in the main scanning direction are preferably greater than 0 mm and smaller than 5 mm.

Now, a description is given of some or all of the advantages according to the embodiment described above, the enumeration of which is not exhaustive or limiting.

According to the embodiment described above, when the end fences 25L and 25R, which are movable in the main scanning direction, overlap the binding positions, the crimp binder 27 crimps and binds the binding positions after the end fences 25L and 25R are retracted. Thus, the crimp binder 27 crimps and binds selected positions in the main scanning direction on the sheet bundle Pb. Since a preferred number of binding positions secured, a preferred binding strength is maintained for the sheet bundle Pb.

According to the embodiment described above, the binding strength is increased as illustrated in FIG. 12 by the crimp binding at a plurality of binding positions at given intervals in the main scanning direction. Thus, as compared with a case where the intervals between the binding positions are 0, a preferred binding strength is obtained for the sheet bundle Pb even when the number of binding positions is reduced.

According to the embodiment described above, by changing the intervals between the binding positions in the main scanning direction depending on the width of the sheet P in the main scanning direction, a preferred number of binding positions is determined for the sheet bundles Pb in various sizes. As a result, a preferred binding strength is obtained for the sheet bundle Pb regardless of the size of the sheet bundle Pb.

According to the embodiment described above, before the end fences 25L and 25R are moved to positions where the end fences 25L and 25R do not overlap the binding positions, the crimp binder 27 crimps and binds at least one binding position, to prevent the misalignment of the sheet brindle Pb that may be caused by the movement of the end fences 25L and 25R.

In the case where the end fences 25L and 25R and the crimp binder 27 are respectively driven by the end-fence motors 38L and 38R and the crimper movement motor 238 that are separate from each other as illustrated in FIG. 6A, the positions and timings of movement of the end fences 25L and 25R and the crimp binder 27 are individually optimized. On the other hand, in the case where the end fences 25L and 25R and the crimp binder 27 are driven by the common crimper movement motor 238 as illustrated in FIG. 6B, cost reduction and space saving of the post-processing apparatus 3 are achieved.

Now, a description is given of a first modification of the embodiment described above.

Specifically, referring now to FIGS. 13A to 13C, a description is given of an operation of the post-processing apparatus 3 according to the first modification.

FIGS. 13A to 13C are diagrams illustrating the positions of the side fences 26L and 26R during a binding process according to the first modification.

Note that detailed descriptions will be omitted of common features of the embodiment described above and the present

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modification. The following description concentrates on the differences between the embodiment described above and the present modification.

According to the first modification, the controller 100 moves the side fences 26L and 26R so that the interval between the side fences 26L and 26R is greater than the width of the Sheet P in the main scanning direction as illustrated in FIG. 13A when the sheet P is accommodated in the internal tray 22. Thus, the sheet P smoothly enters the internal tray 22. On the other hand, according to the first modification, the controller 100 moves the side fences 26L and 26R so that the interval between the side fences 26L and 26R is smaller than the width of the sheet P in the main scanning direction as illustrated in FIG. 13B, before retracting the end fences 25L and 25R in step S505 of FIG. 8 (in other words, before moving the end fences 25L and 25R in the main scanning direction). In other words, according to the first modification, the controller 100 retracts the end fences 25L and 25R in step S505 of FIG. 8 while the interval between the side fences 26L and 26R is smaller than the width of the sheet P in the main scanning direction as illustrated in FIG. 13C.

According to the first modification, the controller 100 moves the end fences 25L and in the main scanning direction while the sheet bundle Pb is fixed to the internal tray 22 by the side fences 26L and 26R, to prevent the position of the sheet bundle Pb from being shifted due to the movement of the end fences 25L and 25R.

Since a desired position is crimped and bound by the crimp binder 27, a preferred binding strength is obtained.

Now, a description is given of a second modification of the embodiment described above.

Specifically, referring now to FIGS. 14A and 14B, a description is given of an operation of the post-processing apparatus 3 according to the second modification. FIGS. 14A and 14B are diagrams illustrating the positions of the tapping roller 23 during a binding process according to the second modification.

Note that detailed descriptions will be omitted of common features of the embodiment described above and the present modification. The following description concentrates on the differences between the embodiment described above and the present modification.

According to the second modification, the controller 100 retracts the tapping roller 23 above the internal tray 22 as illustrated in FIG. 14A when the sheet P sandwiched between the rollers of the conveyance roller pair 15 enters the internal tray 22. According to the second modification, the controller 100 brings the tapping roller 23 into contact with the upper face of the sheet P that has entered the internal tray 22, to separate the sheet P from the conveyance roller pair 15. Thus, the sheet P conveyed by the conveyance roller pair 15 is smoothly accommodated in the internal tray 22.

According to the second modification, the controller 100 brings the tapping roller 23 into contact with the uppermost sheet P placed on the internal tray 22 (in other words, the uppermost sheet P of the sheet bundle Pb) as illustrated in FIG. 14B, before retracting the end fences 25L and 25R in step S505 of FIG. 8. In other words, according to the second modification, the controller 100 retracts the end fences 25L and 25R in step S505 of FIG. 8 while the tapping roller 23 is in contact with the uppermost sheet P placed on the internal tray 22.

According to the second modification, the controller 100 moves the end fences 25L and 25R in the main scanning direction while the sheet bundle Pb is fixed to the internal

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tray 22 by the tapping roller 23, to prevent the position of the sheet bundle Pb from being shifted due to the movement of the end fences 25L and 25R. Since a desired position is crimped and bound by the crimp binder 27, a preferred binding strength is obtained.

Now, a description is given of third and fourth modifications of the embodiment described above.

Specifically, referring now to FIGS. 15A to 16C, a description is given of operations of the post-processing apparatus 3 according to the third and fourth modifications. FIGS. 15A to 15C are diagrams illustrating the positions of the side fences 26L and 26R during a binding process according to the third modification.

FIGS. 16A to 16C are diagrams illustrating the positions of the internal tray 22 and the side fences 26L and 26R during a binding process according to the fourth modification.

Note that detailed descriptions will be omitted of common features of the embodiment described above and the present modifications. The following description concentrates on the differences between the embodiment described above and the present modifications.

According to the third and fourth modifications, in step S504 of FIG. 8, the controller 100 determines whether the end fence 25L or 25R overlaps the binding position N when the internal tray 22 is viewed in the thickness direction and whether the release claw 29 overlaps the binding position N when the internal tray 22 is viewed in the thickness direction. According to the third and fourth modifications, when the controller 100 determines that the release claw 29 overlaps the binding position S3 as illustrated in FIGS. 15A and 16A, the controller 100 moves the sheet bundle Pb in the main scanning direction to a position where the release claw 29 is away from the binding position S3 as illustrated in FIGS. 15B and 16B, before retracting the end fences 25L and 25R in step S505 of FIG. 8. In other words, according to the third and fourth modifications, the controller 100 causes the crimp binder 27 to crimp and bind the sheet bundle Pb in step S506 of FIG. 8 after relatively moving the release claw 29 and the sheet bundle Pb in the main scanning direction so that the release claw 29 is away from the binding position S3, as illustrated in FIGS. 15C and 16C.

According to the third modification illustrated in FIGS. 15A to 15C, the controller 100 moves the side fences 26L and 26R in one of the main scanning directions (rightward in the example of FIGS. 15A to 15C) relative to the internal tray 22. As a result, the sheet bundle Pb is moved in the main scanning direction on the internal tray 22. The relative movement of the release claw 29 to a position away from the binding position S3 allows the crimp binder 27 to crimp and bind the sheet bundle Pb.

On the other hand, according to the fourth modification illustrated in FIGS. 16A to 16C, the controller 100 moves the internal tray 22 and the side fences 26L and 26R together in one of the main scanning directions (rightward in the example of FIGS. 16A to 16C). As a result, the sheet bundle Pb is moved in the main scanning direction together with the internal tray 22. The relative movement of the release claw 29 to a position away from the binding position S3 allows the crimp binder 27 to crimp and bind the sheet bundle Pb.

According to the third and fourth modifications, the binding position is crimped and bound by the crimp binder 27 even when the release claw 29 that does not move in the main scanning direction overlaps the binding position. As a result, a preferred binding strength is obtained for the sheet bundle Pb. According to the third modification, since the side fences 26L and 26R are moved while the internal tray

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22 is fixed, an assembly for moving the internal tray 22 may be omitted. On the other hand, according to the fourth modification, since the internal tray 22, the side fences 26L and 26R, and the sheet bundle Pb are moved together, the sheet bundle Pb is prevented from being bent on the internal tray 22.

The embodiments of the present disclosure are applied to the crimp binder 27 that executes the edge binding as described above. However, the embodiments of the present disclosure may be applied to the saddle binder 33 that executes the saddle binding.

The control method described above may be implemented by, for example, a program. In other words, the control method may be executed by a computer causing an arithmetic device, a storage device, an input device, an output device, and a control device to operate in cooperation with each other based on a program. The program may be written in, for example, a storage device or a storage medium and distributed. Alternatively, the program may be distributed through, for example, an electric communication line.

Now, a description is given of some aspects of the present disclosure. According to a first aspect, a medium processing apparatus includes a conveyor, a receptacle, a conveyance-direction aligner, a crimp binder, and a controller. The conveyor conveys a sheet-shaped medium in a conveyance direction. The receptacle can hold a plurality of media, including the medium, conveyed by the conveyor. The conveyance-direction aligner contacts a downstream end, in the conveyance direction, of the plurality of media placed on the receptacle and aligns a position, in the conveyance direction, of the plurality of media. The crimp binder is disposed to face the downstream end, in the conveyance direction, of the plurality of media placed on the receptacle to press and deform a binding position on the plurality of media of which the position in the conveyance direction is aligned by the conveyance-direction aligner, to bind the plurality of media. The controller controls operations of the conveyor, the conveyance-direction aligner, and the crimp binder. The conveyance-direction aligner and the crimp binder are movable in a main scanning direction orthogonal to each of the conveyance direction and a thickness direction of the medium placed on the receptacle. The controller determines whether the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the controller determines that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction, the controller moves the conveyance-direction aligner in the main scanning direction to a position away from the binding position. The controller then causes the crimp binder to crimp and bind the plurality of media.

According to a second aspect, the medium processing apparatus of the first aspect further includes a pair of conveyance-direction aligners including the conveyance-direction aligner. The pair of conveyance-direction aligners is arranged opposite each other across a center position, in the main scanning direction, of the plurality of media. The controller increases or decreases an interval between the pair of conveyance-direction aligners while maintaining the pair of conveyance-direction aligners at equal distances from the center position.

According to a third aspect, in the medium processing apparatus of the first or second aspect, the controller causes the crimp binder to sequentially crimp and bind a plurality of binding positions, including the binding position, arranged at given intervals in the main scanning direction.

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According to a fourth aspect, in the medium processing apparatus of the third aspect, the controller changes the intervals between the plurality of binding positions depending on a width of the plurality of media in the main scanning direction.

According to a fifth aspect, in the medium processing apparatus of the third or fourth aspect, the controller causes the crimp binder to crimp and bind at least one binding position of the plurality of binding positions, the at least one binding position being away from the conveyance-direction aligner. The controller then moves the conveyance-direction aligner in the main scanning direction.

According to a sixth aspect, in the medium processing apparatus of any one of the third to fifth aspects, the intervals between the plurality of binding positions are greater than 0 mm and less than 5 mm.

According to a seventh aspect, the medium processing apparatus of any one of the first to sixth aspects further includes a releaser that contacts the downstream end, in the conveyance direction, of the plurality of media and moves upstream in the conveyance direction to release the plurality of media crimped and bound from the receptacle. The controller determines whether the releaser overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the controller determines that the releaser overlaps the binding position when the receptacle is viewed in the thickness direction, the controller relatively moves the releaser and the plurality of media to cause the releaser to be away from the binding position. The controller then causes the crimp binder to crimp and bind the binding position.

According to an eighth aspect, the medium processing apparatus of the seventh aspect further includes a pair of width-direction aligners that contacts opposed ends, in the main scanning direction, of the plurality of media and moves in the main scanning direction to align a position, in the main scanning direction, of the plurality of media. The controller determines whether the releaser overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the controller determines that the releaser overlaps the binding position when the receptacle is viewed in the thickness direction, the controller moves the pair of width-direction aligners in the main scanning direction relative to the receptacle to move the plurality of media on the receptacle in the main scanning direction.

According to a ninth aspect, the medium processing apparatus of the seventh aspect further includes a pair of width-direction aligners that contacts opposed ends, in the main scanning direction, of the plurality of media and moves in the main scanning direction to align a position, in the main scanning direction, of the plurality of media. The controller determines whether the releaser overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the controller determines that the releaser overlaps the binding position when the receptacle is viewed in the thickness direction, the controller moves the receptacle and the pair of width-direction aligners together in the main scanning direction to move the plurality of media together with the receptacle in the main scanning direction.

According to a tenth aspect, in the medium processing apparatus of any one of the first to ninth aspects, the controller moves the conveyance-direction aligner in the main scanning direction to a position away from the binding position while the plurality of media is fixed to the receptacle.

According to an eleventh aspect, the medium processing apparatus of the tenth aspect further includes a pair of

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width-direction aligners that contacts opposed ends, in the main scanning direction, of the plurality of media and moves in the main scanning direction to align a position, in the main scanning direction, of the plurality of media. The controller moves the conveyance-direction aligner in the main scanning direction while an interval between the pair of width-direction aligners is smaller than a width of the medium in the main scanning direction.

According to a twelfth aspect, the medium processing apparatus of the tenth aspect further includes a contact and separation device that comes into contact with and separates from an uppermost surface of the plurality of media. The controller moves the conveyance-direction aligner in the main scanning direction while the contact and separation device is in contact with the uppermost surface of the plurality of media.

According to a thirteenth aspect, the medium processing apparatus of any one of the first to twelfth aspects further includes a drive source and a driving force transmission assembly that moves the conveyance-direction aligner and the crimp binder in the main scanning direction by the driving source.

According to a fourteenth aspect, the medium processing apparatus of any one of the first to twelfth aspects further includes a first driving source that moves the conveyance-direction aligner in the main scanning direction and a second driving source that moves the crimp binder in the main scanning direction.

According to a fifteenth aspect, an image forming system comprising an image forming apparatus and the medium processing apparatus of any one of the first to fourteenth aspects. The image forming apparatus forms an image on a medium. The medium processing apparatus crimps and binds the plurality of media on each of which the image is formed by the image forming apparatus.

According to a sixteenth aspect, a medium processing method includes determining whether a conveyance-direction aligner overlaps a binding position when a receptacle is viewed in a thickness direction of a medium, moving the conveyance-direction aligner in a main scanning direction to a position away from the binding position based on a determination that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction of the medium, the main scanning direction being a direction orthogonal to each of a conveyance direction of the medium and the thickness direction of the medium placed on the receptacle, and moving crimp binder in the main scanning direction and causing the crimp binder to crimp and bind a plurality of media including the medium.

According to a seventeenth aspect, a non-transitory recording medium stores a plurality of instructions which, when executed by one or more processors, causes the processors to perform a medium processing method. The method includes determining whether a conveyance-direction aligner overlaps a binding position when a receptacle is viewed in a thickness direction of a medium, moving the conveyance-direction aligner in a main scanning direction to a position away from the binding position based on a determination that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction of the medium, the main scanning direction being a direction orthogonal to each of a conveyance direction of the medium and the thickness direction of the medium placed on the receptacle, and moving crimp

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binder in the main scanning direction and causing the crimp binder to crimp and bind a plurality of media including the medium.

According to one aspect of the present disclosure, selected positions in a main scanning direction on a bundle of media placed on a tray can be crimped and bound, and thus a preferred binding strength can be obtained.

The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention. It is therefore to be understood that the disclosure of this patent specification may be practiced otherwise by those skilled in the art than as specifically described herein and such modifications and alternatives are within the technical scope of the appended claims.

Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.

The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose, processors, integrated circuits, application specific integrated circuits (ASICs), digital signal processors (DSPs), field programmable gate arrays (FPGAs), conventional circuitry and/or combinations thereof which are configured or programmed to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carry out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein or otherwise known which is programmed or configured to carry out the recited functionality. When the hardware is a processor which may be considered a type of circuitry, the circuitry, means, or units are a combination of hardware and software, the software being used to configure the hardware and/or processor.

The invention claimed is:

1. A medium processing apparatus comprising:

a conveyor including a conveyance roller configured to convey a sheet-shaped medium in a conveyance direction;

a tray configured to hold the medium conveyed by the conveyor;

a conveyance-direction fence configured to contact a downstream end, in the conveyance direction, of the medium placed on the tray and aligns a position, in the conveyance direction, of the medium;

a crimp binder disposed to face the downstream end, in the conveyance direction, of the medium placed on the tray to press and deform a binding position on a plurality of media, including the medium, of which the position in the conveyance direction is aligned by the conveyance-direction fence, to bind the plurality of media;

a pair of width-direction fences configured to contact opposed ends, in a main scanning direction, of the plurality of media and move in the main scanning direction to align a position, in the main scanning direction, of the plurality of media, the main scanning direction being orthogonal to each of the conveyance direction and a thickness direction of the medium placed on the tray; and

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circuitry configured to control operations of the conveyor, the conveyance-direction fence, and the crimp binder, the conveyance-direction fence and the crimp binder being movable in the main scanning direction, the circuitry being configured to:

determine whether the conveyance-direction fence overlaps the binding position when the tray is viewed in the thickness direction;

move the conveyance-direction fence in the main scanning direction to a position away from the binding position while the plurality of media is fixed to the tray, in a case where the circuitry determines that the conveyance-direction fence overlaps the binding position when the tray is viewed in the thickness direction; and

cause the crimp binder to crimp and bind the plurality of media, and

the circuitry being further configured to move the conveyance-direction fence in the main scanning direction while the opposed ends of the plurality of media contact the pair of width-direction fences.

2. The medium processing apparatus according to claim 1, further comprising a pair of conveyance-direction fences including the conveyance-direction fence, the pair of conveyance-direction fences arranged opposite each other across a center position, in the main scanning direction, of the plurality of media,

wherein the circuitry is configured to increase or decrease an interval between the pair of conveyance-direction fences while maintaining the pair of conveyance-direction fences at equal distances from the center position.

3. The medium processing apparatus according to claim 1, wherein the circuitry is configured to cause the crimp binder to sequentially crimp and bind a plurality of binding positions, including the binding position, arranged at given intervals in the main scanning direction.

4. The medium processing apparatus according to claim 3, wherein the circuitry is configured to change the intervals between the plurality of binding positions depending on a width of the plurality of media in the main scanning direction.

5. The medium processing apparatus according to claim 3, wherein the circuitry is configured to cause the crimp binder to crimp and bind at least one binding position of the plurality of binding positions, the at least one binding position being away from the conveyance-direction fence, and move the conveyance-direction fence in the main scanning direction.

6. The medium processing apparatus according to claim 3, wherein the intervals between the plurality of binding positions are greater than 0 mm and less than 5 mm.

7. The medium processing apparatus according to claim 1, further comprising a contact configured to contact the downstream end, in the conveyance direction, of the plurality of media and move upstream in the conveyance direction to release the plurality of media crimped and bound from the tray,

wherein the circuitry is configured to:

determine whether the contact overlaps the binding position when the tray is viewed in the thickness direction;

relatively move the contact and the plurality of media to cause the contact to be away from the binding position in a case where the circuitry determines that the contact overlaps the binding position when the tray is viewed in the thickness direction; and

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cause the crimp binder to crimp and bind the binding position.

8. The medium processing apparatus according to claim 7, wherein the circuitry is configured to:

determine whether the contact overlaps the binding position when the tray is viewed in the thickness direction; and

move the pair of width-direction fences in the main scanning direction relative to the tray to move the plurality of media on the tray in the main scanning direction, in a case where the circuitry determines that the contact overlaps the binding position when the tray is viewed in the thickness direction.

9. The medium processing apparatus according to claim 7, wherein the circuitry is configured to:

determine whether the contact overlaps the binding position when the tray is viewed in the thickness direction; and

move the tray and the pair of width-direction fences together in the main scanning direction to move the plurality of media together with the tray in the main scanning direction, in a case where the circuitry determines that the contact overlaps the binding position when the tray is viewed in the thickness direction.

10. The medium processing apparatus according to claim 1, further comprising a contact and separation device includ-

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ing a tapping roller configured to come into contact with and separate from an uppermost surface of the plurality of media,

wherein the circuitry is configured to move the conveyance-direction fence aligner in the main scanning direction while the contact and separation device is in contact with the uppermost surface of the plurality of media.

11. The medium processing apparatus according to claim 1, further comprising:

a drive source; and

a driving force transmission assembly including a pulley and a timing belt configured to move the conveyance-direction fence and the crimp binder in the main scanning direction by the driving source.

12. The medium processing apparatus according to claim 1, further comprising:

a first driving source including a first motor configured to move the conveyance-direction fence in the main scanning direction; and

a second driving source including a second motor configured to move the crimp binder in the main scanning direction.

13. An image forming system comprising:

the medium processing apparatus according to claim 1, the medium processing apparatus being configured to crimp and bind the plurality of media on each of which an image is formed.

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